Highlights

Certified Correction of Object Hallucinations in Vision–Language Models:
Guarantees, Gains and Failure Modes under a Pre-specified Protocol

Quang-Vinh Dang

- A repair-and-certify selective predictor with a valid finite-sample risk guarantee
- Proved and evaluated procedures coincide: fixed value grid, image-grouped folds
- Certified gain over filtering: +0.8 to +7.3 points; CI excludes zero in 7 of 12
- Much repair is detector confirmation; false positives are rarely corrected
- A small matched-prefix test suggests mid-caption repair raises hallucination

Certified Correction of Object Hallucinations in Vision–Language Models: Guarantees, Gains and Failure Modes under a Pre-specified Protocol

Quang-Vinh Dang[a,*]

[a] *British University Vietnam, Hung Yen, Vietnam*

Abstract

Selective prediction gives a vision–language model (VLM) a distribution-free guarantee on its emitted answers at the price of abstaining on the rest. When the base error rate μ exceeds the target α, any predictor that only emits or abstains the model's own answer must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs. For object-existence questions we study a predictor that may emit a different answer: Certified-Correction Risk Control (CCRC) accepts the model's answer where a correctness score is high, substitutes the answer of a distinct evidence source where that source is decisive, abstains otherwise, and certifies the error rate of everything emitted with a fixed-sequence Clopper–Pearson test. We prove validity for policy families indexed by pre-specified score values under i.i.d. sampling of calibration units, and evaluate under a protocol in which the proved and evaluated procedures coincide: folds are unions of images, and thresholds, repair gate and sequence start are fixed before calibration. On six benchmark–backbone cells, certified coverage over filtering rises by 0.8 to 7.3 points with a gate fixed on external data at the target's calibration size, positive in all twelve combinations; an image-level bootstrap excludes zero in seven. Most repaired items re-emit the model's own answer with detector confirmation. The correctness score supplies most of the coverage; fixed-sequence testing is fragile

*Corresponding author. ORCID: 0000-0002-3877-8024
Email address: vinh.dq4@buv.edu.vn (Quang-Vinh Dang)

with small calibration folds. A small matched-prefix experiment suggests that repairing a claim mid-caption increases downstream hallucination.

Keywords: conformal prediction, selective prediction, risk control, vision–language models, hallucination, uncertainty quantification

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1. Introduction

Hallucinated object claims limit the use of vision–language models (VLMs) wherever a wrong answer is costly. Selective prediction addresses this by letting the model abstain: a score ranks the model's answers, a threshold is calibrated on held-out data, and the model answers only above it. Conformal and risk-controlling variants [1, 2, 3] certify, with a finite-sample guarantee that assumes nothing about the score or the model, that the error rate among emitted answers stays below a target α. Existing conformal methods for VLMs filter, back off or abstain [4, 5, 6]; mitigation methods repair outputs but give no guarantee [7, 8, 9, 10].

The guarantee has a price that no score can remove. Kotte [11] shows that when the base error rate μ exceeds α, any predictor that only emits or abstains the model's own answer must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs (Proposition 1). The proof uses one premise, that an abstained item's model answer stays wrong, so a predictor that may emit a *different* answer lies outside it.

We study that possibility for yes/no object-existence questions, with an open-vocabulary detector as a second, distinct source of evidence. *Certified-Correction Risk Control* (CCRC) accepts the model's answer where a correctness score is high, replaces it with the detector's answer where the detector's evidence is decisive, abstains otherwise, and certifies the error rate of the whole emitted set, accepted and substituted answers together (Figure 1). The scope is deliberate: existence claims are the hallucinations a detector can refute, and §9 discusses what other evidence channels would be needed for attributes, counts and relations.



Figure 1: The three actions of CCRC on real POPE images (LLaVA-1.5, OWLv2). *Accept*: the correctness score is high and the model is right. *Repair*: the model misses an object that is present and scores low, the detector finds it with high confidence, and its answer is emitted instead. *Abstain*: neither source is reliable (here both are wrong with weak evidence), so no answer is emitted.

What is and is not new. Theorem 1 is an application of fixed-sequence testing [2] with exact binomial bounds, and we claim no new statistical technique. The contributions are the formulation (a risk certificate over a mixture of the model's and a second predictor's answers, which escapes the abstention floor), a protocol under which the proved procedure is the evaluated procedure, and measurements of what the mechanism does and does not contribute. Four design decisions make up the protocol.

1. *The proved procedure is the evaluated procedure.* Theorem 1 covers policy families indexed by threshold *values* fixed before calibration, under i.i.d. sampling of calibration units. Thresholds and repair gate are read off the fit fold, never off the calibration sample whose errors are counted.
2. *The sampling unit is respected.* POPE contributes six questions per image.

Folds are unions of whole images, the within-image dependence of errors is measured, and a known-population simulation shows how clustering would degrade the guarantee and how one-item-per-image calibration restores it.

3. *The repair gate is fixed without looking at the evaluation data.* Three ways keep the guarantee: a constant chosen on a different benchmark at the target's calibration size, a choice made on the fit fold, and a union bound over a pre-specified set of gates.
4. *Uncertainty refers to the data.* Gains carry an outer image-level bootstrap interval, since repeated random splits of one benchmark measure sensitivity to the split and say nothing about the benchmark's sampling error.

What we find.

- With the gate fixed on an external benchmark at the target's calibration size, CCRC certifies +0.8 to +7.3 points more coverage than filtering in all twelve (cell, α) combinations, and the image-bootstrap interval excludes zero in seven. The gain is about one point on cells with 150–200 calibration items and up to +7.3 points with 500 or more (§6.1).

- Choosing the gate on the fit fold, or certifying three gates by a union bound, is valid and helps with 500 or more calibration items (+2.4 to +8.8 and +2.0 to +4.4 points) but costs more than it returns with 150–200 (-6.2 to +9.4 and -16.5 to +4.4 points).

- Much of the repair is confirmation: 22–100% of repaired items are items on which the detector agrees with the model. With a small gate the answer is changed for under 0.5% of test items; with a large gate for 2.4–5.7%, of which 81–93% are right, and 4–5% of the model's false-positive errors are corrected (§6.3).

- The start of the fixed sequence matters more than the repair on small folds. Starting where a single error can be tolerated ($k \geq 38$ at $\alpha = \delta =$

0.10) rather than where none can ($k \geq 22$) raises filtering coverage on the smallest cell from 22% to 41% and removes an apparent loss of repair on AMBER, where the conventional start makes CCRC abort on 72% of splits (§6.4).

- Certified coverage comes mostly from the score: a detection-grounded score certifies 68% on POPE-1500 against 41% for model confidence and 0.5% for the image–text similarity score of ConfLVLM, while certified repair adds 1.4 to 8.8 points (§6.5).
- In free-form generation, a small matched-prefix experiment ($n = 121$, under-powered for the size of the effect) finds more downstream hallucination after repair (§8); it withdraws support for a fixed-point argument for a sequential guarantee.

Scope. Everything here concerns binary existence questions, one detector, one family of backbones and benchmarks whose images overlap (1,457 distinct images across all cells: the 500 POPE images, nested across the POPE cells, and 957 AMBER images). We do not claim that correction is generally useful, only that under a protocol whose guarantee we can state it is valid and its gain is measured.

2. Related work

The survey covers work available up to 4 October 2026; claims of the form "no method does X" refer to the literature we surveyed up to that date and are not claims about the field.

Hallucination in vision–language models. Multimodal hallucination is usually split into existence, attribute and relation errors [12]; surveys [13, 14, 15] attribute it to language-prior dominance, weak visual attention, instruction-tuning artefacts [16] and miscalibration [17]. Existence claims can be checked by an open-vocabulary detector [18, 19]; attribute and relation claims need finer perception than a detector supplies, which is why the benchmark's composition,

and not only its difficulty, determines whether our evidence channel applies (§6.7). POPE [20] probes existence with balanced yes/no questions under random, popular and adversarial negative sampling; AMBER [12] adds an LLM-free discriminative split; MME [21], GQA [22] and HallusionBench [23] are used here as settings where the detector mostly does not apply.

Mitigation methods modify decoding [24, 25, 26, 27, 28, 29], intervene on heads or positional structure [30, 31, 32, 33, 34], retrain [35, 36, 16], or verify and rewrite after generation [7, 8, 9, 37, 38, 10]. Post-hoc verification performs the same three steps as CCRC (flag, consult external evidence, substitute) and none of these systems certifies the substitution. Lowering the base risk μ lowers the abstention floor, so these methods compose with ours; §6.5 runs CCRC on top of visual contrastive decoding.

Conformal prediction and distribution-free risk control. Conformal prediction [39, 40, 41, 42] gives finite-sample coverage under exchangeability. Risk-controlling prediction sets [1] and conformal risk control [3] control the expectation of a monotone loss; Learn-then-Test [2] treats the choice among configurations as multiple testing and admits fixed-sequence testing over a pre-ordered family at full δ, which we use. For a binary loss the exact Clopper–Pearson bound [43] is tighter than Hoeffding, empirical Bernstein or betting bounds [44]. Selective prediction with abstention predates the conformal treatment [45, 46, 47, 48, 49]; the floor of Proposition 1 concerns the action space and applies to these formulations too.

Conformal methods for language and vision–language models. Conformal language modelling [50] calibrates sampling, stopping and rejection; conformal factuality [51] backs off to a subset of claims; coherent factuality [52] filters whole reasoning steps; adaptive variants improve conditional coverage of the filter [53]; and a robustness audit shows that claim filtering for retrieval-augmented models becomes vacuous at high factuality levels and is fragile to shift [54]. Conformal alignment [55], conformal multiple choice [56] and calibrated help-seeking [57] certify act-or-defer decisions. In the multimodal setting ConfLVLM [4] filters

unsupported claims scored by image–text similarity, Wagner et al. [58] study abstention along correctness and answerability, and IntroConformal [6] builds conformity scores from the model's hidden states and self-verification, without external verifiers. Inference-time conformal reasoning [59] moves calibration inside generation and intervenes by truncating. All of these filter, back off, stop or abstain. One method certifies a revised output: conformal linguistic calibration [60] replaces claims by less specific ones with a guarantee on the rewritten text; it is text-only and the rewrite generalises a claim rather than substituting externally verified evidence.

Learning to defer, cascades and routing. Emitting the answer of a second predictor is the action of learning to defer, where a policy decides when a decision-maker or expert answers instead of the model [61, 62]; classical rejection learning [48] abstains at a constant cost. Cascades and routing send an input to a cheaper or stronger model by confidence [63], and several recent methods add statistical guarantees: cascaded selective evaluation certifies agreement with human judgments by fixed-sequence testing over LLM judges [64], conformal cascades and edge–cloud alignment control coverage or false discoveries across model tiers [65, 66], conformal arbitrage calibrates when a guardian model is consulted [67], and conformal deferral routes to human experts without training [68]. Cascaded selective evaluation is the closest precedent for a finite-sample certificate over an emitted mixture, and it shares the fixed-sequence machinery. The settings differ in what the second predictor is. In those works it is a stronger model of the same kind, a human, or an expert whose answer is requested by the system's own confidence; here it is a detector of a different modality whose evidence margin, not the VLM's confidence, gates the substitution, and there is no learned deferral policy. Our reading is that a stronger model of the same kind tends to share the weaker model's failure modes, whereas a detector may err on different items, which is the property that makes substitution useful and that Section 6.9 probes by removing the detector from the score.

Uncertainty estimation without a guarantee. Our correctness score belongs to the literature on generative uncertainty: self-evaluation [69], semantic entropy [70, 71], verbalised confidence [72], sampling consistency [73, 74], and token-level conformal variants [75, 76]. These scores interrogate the model about its own output; detector and similarity scores are perceptual evidence from a distinct predictor.

The closest works. Kotte [11] proves the abstention floor for text-only structured generation and recommends relaxing α; the floor admits three responses: relax the target, lower μ or sharpen the score, or change the action space. We take the third. BCEA [5] sharpens the score by spending a visual budget to re-read the image and recalibrating on post-acquisition scores; the answer is still the model's own. Its current version certifies a Bonferroni-corrected Clopper–Pearson test over a threshold grid fixed independently of the calibration sample, and labels scanning the calibration scores at full confidence an uncertified practical variant. CEBC [77] is the closest published method that edits unsupported object mentions in VLM output with an external detector: a detector-score threshold is set as a quantile of the scores of hallucinated mentions in base captions, a VLM rewrites unsupported mentions, and a final filter deletes what remains. It states no theorem, assumes no sampling model and does not place the revised text under a calibrated guarantee. §7 compares the three in detail.

Shift caused by the predictor. Editing a generation changes what follows. This is feedback covariate shift [78], to which weighted conformal prediction [79] does not apply; online adaptation [80, 81] and reweighting [82] address parts of it. §8 shows that in generative correction the obstacle is stronger than loss of exchangeability.

What is new here. A selective predictor whose action space includes substitution and whose certificate covers the emitted mixture; a protocol in which the proved procedure and the evaluated procedure coincide; and measurements that bound what the mechanism contributes: attribution against score quality, accounting

of what a repair changes, the fragility of the fixed sequence at small calibration sizes, and the failure of mid-sequence repair.

3. Problem setup

3.1. Units, risk and the guarantee

Let $Z = (X, A^\star)$ be a query–answer pair, where X contains an image and a question and $A^\star \in \{0, 1\}$ is the gold answer. A VLM returns $\hat{a}(X)$. A second predictor (channel 2) returns an answer $b(X)$ and an evidence margin $m(X) \geq 0$. Write $Y = \mathbf{1}\{\hat{a}(X) = A^\star\}$ and $Y^{(2)} = \mathbf{1}\{b(X) = A^\star\}$ for the two correctness indicators and $\mu = \mathbb{P}[Y = 0]$ for the base risk. α is always a risk target and δ the failure probability of the certificate.

A *policy* π maps X to one of three actions: emit $\hat{a}(X)$, emit $b(X)$, or abstain. Let $\mathcal{E}_\pi(X) = \mathbf{1}\{\pi \text{ emits}\}$ be the emission indicator and $W_\pi = \mathcal{E}_\pi \cdot \mathbf{1}\{\text{emitted answer} \neq A^\star\}$ the indicator of an emitted error. The *risk* and *coverage* of π are

$$\mathcal{R}(\pi) = \frac{\mathbb{E}[W_\pi]}{\mathbb{P}[\mathcal{E}_\pi = 1]}, \qquad \mathcal{C}(\pi) = \mathbb{P}[\mathcal{E}_\pi = 1],$$

with the convention $\mathcal{R}(\pi) = 0$ when $\mathbb{P}[\mathcal{E}_\pi = 1] = 0$. The same convention applies to a realised abort: a procedure that certifies nothing emits nothing, has risk 0 and coverage 0, and every coverage figure in this paper includes aborted splits as zero coverage.

Assumption 1 (Sampling unit)**.** The calibration units $Z_1, \ldots, Z_n$ are i.i.d. draws from a distribution P, independent of the data used to build the policy family (the *fit* data), and the unit is the item.

Exchangeability is not enough here. Under exchangeability alone, sampling without replacement from a finite pool makes the number of emitted errors hypergeometric rather than binomial, and the exact binomial bound used below is not valid. When several questions share an image the item is not an i.i.d. unit; §4.4 states what remains true and measures what does not.

Assumption 2 (Pre-specified policy family)**.** The candidate policies form a finite, ordered family $\pi_1, \ldots, \pi_M$ whose members and order are functions of the fit data and of constants only, never of calibration outcomes or calibration features.

Calibration is a statement about the *population* risk $\mathcal{R}(\pi_{\hat{\jmath}})$ of the policy the calibration sample selects; it is not a statement about any finite test fold, whose realised error rate carries binomial noise of its own.

3.2. Filtering and the floor it cannot cross

A *filtering* policy uses a score $s(X) \in [0, 1]$, large when the model is likely right, and emits $\hat{a}(X)$ when $s(X) \geq t$.

Proposition 1 (Abstention floor; Kotte [11])**.** *Let $\mu > \alpha$. Any selective predictor whose emitted answer is always $\hat{a}(X)$ and whose risk is at most α satisfies*

$$1 - \mathcal{C} \;\geq\; \frac{\mu - \alpha}{1 - \alpha}.$$

Proof. Split the error mass by emission, $\mu = \mathcal{R}\,\mathcal{C} + \mathbb{P}[Y = 0 \mid \text{abstain}](1 - \mathcal{C}) \leq \alpha\mathcal{C} + (1 - \mathcal{C})$. The inequality uses that abstaining does not change Y. Solving for $\mathcal{C}$ gives $\mathcal{C} \leq (1 - \mu)/(1 - \alpha)$. □

The proof uses one premise: an abstained item's model answer is still wrong. It is a statement about the action space. Because α is chosen by the user and μ is estimated from data, the floor can be evaluated before calibration only up to the uncertainty of $\hat{\mu}$; a feasibility check should use a confidence bound on μ, not its plug-in value.

3.3. Two channels

Definition 1 (Channels)**.** *Channel 1* is a correctness score $s(X) \approx \mathbb{P}[Y = 1 \mid \phi(X)]$ fitted on features of the model's own output together with grounding features. *Channel 2* is a distinct predictor $b(X)$ that answers the same question without using the VLM's answer, together with a margin $m(X)$ measuring how decisively it does so.

Table 1: Notation.

α, δ	risk target; failure probability of the certificate
μ	base risk $\mathbb{P}[Y = 0]$
$Y,\ Y^{(2)}$	correctness indicators of the model and of channel 2
$s(X),\ m(X)$	channel-1 score; channel-2 evidence margin
$t_1 > \cdots > t_M$	acceptance thresholds, score *values* fixed from fit data
g	repair-gate value on m, fixed from fit or external data
$\mathcal{E}_j,\ K_j,\ V_j$	emission indicator, emitted count and emitted-error count of policy j on the calibration sample
$U(e, k; \delta)$	one-sided Clopper–Pearson upper bound
$k_{\text{start}}(e_0)$	smallest k with $U(e_0, k; \delta) \leq \alpha$

We call channel 2 a *distinct evidence source*. Statistical independence between Y and $Y^{(2)}$ is neither assumed nor needed for validity: Theorem 1 holds for any fixed policy and needs only the sampling model. What distinctness buys is usefulness. A repair region helps only if $\mathbb{P}[Y^{(2)} = 1 \mid \text{region}]$ is high, which requires channel 2's errors to differ from the model's inside that region (§6.6 shows what happens when channel 2 is the negation of the model's answer). Our instantiation also violates independence in the other direction: the detector score is an input to channel 1.

Instantiation. Channel 2 is OWLv2 [18] queried with the object named in the question. With detection confidence $o(X)$ and a fixed operating point $\theta = 0.15$, $b(X) = \mathbf{1}\{o(X) \geq \theta\}$ and $m(X) = |o(X) - \theta|$. A confident negative reaches only $m \leq \theta$ while positives reach $1 - \theta$, so with a single gate on m every repair says "the object is present". Section 6.3 evaluates a polarity-symmetric gate as well. When no detector value exists (the object phrase could not be parsed, or the question concerns an attribute or relation), the item keeps channel 1's decision, receives $m = -1$, and can never be repaired.

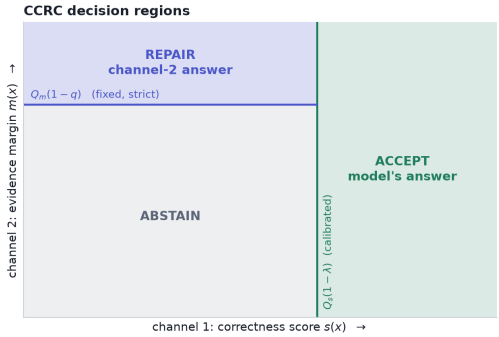


Figure 2: The (s, m) plane under policy (1). The acceptance threshold moves along the sequence; the repair gate is fixed, so the accepted region grows while the repair region shrinks.

4. Certified-correction risk control

4.1. The policy family

Fix a gate value g and a descending grid of acceptance thresholds $t_1 > \cdots > t_M$ on the channel-1 score. Policy π_j acts as

$$\pi_j(x) = \begin{cases} \text{ACCEPT } \hat{a}(x), & s(x) \geq t_j, \\ \text{REPAIR WITH } b(x), & s(x) < t_j \text{ and } m(x) \geq g, \\ \text{ABSTAIN}, & \text{otherwise.} \end{cases} \tag{1}$$

Figure 2 shows the induced partition of the (s, m) plane. The emitted set of π_j is the accepted set together with the repaired set, and an emitted error is a wrong accepted answer or a wrong channel-2 answer: with $A_j = \{s \geq t_j\}$ and $R_j = \{s < t_j,\ m \geq g\}$,

$$W_j = \mathbf{1}\{X \in A_j\}(1 - Y) + \mathbf{1}\{X \in R_j\}(1 - Y^{(2)}). \tag{2}$$

Because g does not depend on j, the emitted set $A_j \cup \{m \geq g\}$ grows as t_j falls. Filtering is the case $g = \infty$.

4.2. How thresholds and gate are fixed

Both are values, and both are computed from the fit fold, which is disjoint from the calibration fold and from the test fold.

- *Score grid.* The channel-1 model is fitted on the fit fold. Its out-of-fold scores on that fold (image-grouped five-fold cross-fitting inside the fit fold) define the grid $t_j = Q_{1-\lambda_j}$ of their empirical distribution, with $\lambda_1 < \cdots < \lambda_{40} = 1$ equally spaced. Using out-of-fold scores keeps the quantile-to-value map faithful to out-of-sample behaviour; an in-sample fit is mildly overconfident on its own data.

- *Start.* The first hypothesis is where $c\,k_{\mathrm{start}}(e_0)$ items are expected to be emitted, $\lambda_1 = \min\{0.9, \max\{0.05,\, c\,k_{\mathrm{start}}(e_0)/n_{\mathrm{cal}}\}\}$, with $k_{\mathrm{start}}(e_0) = \min\{k : U(e_0, k; \delta) \le \alpha\}$ the smallest emitted count at which e_0 observed errors still certify the target (Lemma 1). The default is $e_0 = 1$, $c = 1.5$: start where one error can be tolerated, with half as much again as headroom for fold-to-fold variation in K. The start is computed from α, δ and n_{cal} alone, before any label is read; §6.4 ablates it.

- *Repair gate.* $g = Q_{1-q}(m)$ over grounded fit-fold items. Three ways of fixing q keep the guarantee: (a) *external*: q is chosen once on a different benchmark by running the same procedure there and keeping the candidate (filtering, or $q \in \{0.05, 0.10, 0.25, 0.50\}$) that certifies the most coverage; (b) *fit-fold*: the same choice is made inside each split on the fit fold alone; (c) *union*: three gates $\{0.05, 0.10, 0.25\}$ are each certified at $\delta/3$ and the certified policy with the largest calibration coverage is emitted.

4.3. Validity

Theorem 1 (CCRC validity)**.** *Under Assumptions 1 and 2, let $\hat{\jmath}$ be the last index passed by the fixed-sequence test that proceeds through $j = 1, 2, \ldots$ while $U(V_j, K_j; \delta) \le \alpha$ on the calibration sample and stops at the first failure ($\hat{\jmath} = 0$ if the first test fails, in which case nothing is emitted). Then*

$$\mathbb{P}\big[\mathcal{R}(\pi_{\hat{\jmath}}) \le \alpha\big] \;\ge\; 1 - \delta,$$

where the probability is over the calibration sample, conditional on the fit data.

Proof. Condition on the fit data, so the policies are fixed. For policy j let $K_j = \sum_i \mathcal{E}_j(X_i)$ and $V_j = \sum_i W_j(Z_i)$, and consider $H_j : \mathcal{R}(\pi_j) > \alpha$. Emission is a function of X alone. By Assumption 1 the units are independent, so conditional on the set of emitted units (equivalently, on the emission pattern), the emitted units are i.i.d. draws from $P(\cdot \mid \mathcal{E}_j = 1)$ and each has error indicator $W_j \sim \text{Bernoulli}(\mathcal{R}(\pi_j))$, independently across units. Hence $V_j \mid \{K_j = k\} \sim \text{Bin}(k, \mathcal{R}(\pi_j))$, and by construction of the Clopper–Pearson bound, $\mathbb{P}[U(V_j, K_j; \delta) \leq \alpha \mid K_j = k] \leq \delta$ whenever $\mathcal{R}(\pi_j) > \alpha$ and $k \geq 1$; for $k = 0$ the bound is $1 > \alpha$. So $\phi_j = \mathbf{1}\{U \leq \alpha\}$ is a level-δ test of H_j.

Let $j_0 = \min\{j : H_j \text{ is true}\}$, if it exists. The sequence reaches an index $\geq j_0$ only if H_{j_0} is rejected, an event of probability at most δ. On the complement, $\hat{\jmath} < j_0$, so every index the procedure can return has $\mathcal{R}(\pi_{\hat{\jmath}}) \leq \alpha$. If no H_j is true the claim is trivial, and $\hat{\jmath} = 0$ emits nothing and has risk 0 by convention. □

The proof uses neither monotonicity of the risk along the sequence nor nesting of the emitted sets; nesting is what makes "the last index that passed" a coverage-maximising return. It also does not use whether W_j refers to the model's answer or to channel 2's, which is why Proposition 1 does not bind: its premise restricts the action space, and the certificate never refers to it.

Corollary 1 (Gate fixed on other data, or chosen on the fit fold)**.** *If g is a constant, is computed from external data independent of the calibration sample, or is computed from the fit data (including by a selection rule applied to the fit data alone), the policy family satisfies Assumption 2 and Theorem 1 holds at the full level δ.*

Corollary 2 (Union over a pre-specified set of gates)**.** *Let $\mathcal{Q}$ be a finite set of gate values fixed in advance. Run the fixed-sequence test for each at level $\delta/|\mathcal{Q}|$ and emit any certified policy, chosen by any rule that may use the calibration sample. Then the emitted policy has risk at most α with probability at least $1-\delta$.*

Proof. By Theorem 1 each family fails with probability at most $\delta/|\mathcal{Q}|$; on the complement of the union of these events every certified policy is compliant. □

Corollaries 1 and 2 do not cover a gate chosen by looking at the calibration or test outcomes of the benchmark under evaluation. Section 6.1 reports the continuity gate $q = 0.10$, which was selected in exploratory runs on these benchmarks, as an *uncertified* reference arm.

Lemma 1 (Minimal certifiable emitted set)**.** *$U(0, k; \delta) = 1 - \delta^{1/k}$, so $e_0 = 0$ errors certify α iff $k \geq k_{\text{start}}(0) = \lceil \ln \delta / \ln(1 - \alpha) \rceil$, which is 22 at $\alpha = \delta = 0.10$. For $e_0 \geq 1$, $k_{\text{start}}(e_0)$ is the smallest k with $\mathbb{P}[\text{Bin}(k, \alpha) \leq e_0] \leq \delta$; at $\alpha = \delta = 0.10$ it is $38, 52, 65$ for $e_0 = 1, 2, 3$, and at $\delta/3$ it is $33, 51, 66, 81$ for $e_0 = 0, \ldots, 3$.*

Proof. $\mathbb{P}[\text{Bin}(k, p) \leq e_0]$ is decreasing in p and equals δ at $p = U(e_0, k; \delta)$, so $U(e_0, k; \delta) \leq \alpha$ iff $\mathbb{P}[\text{Bin}(k, \alpha) \leq e_0] \leq \delta$. For $e_0 = 0$ this is $(1 - \alpha)^k \leq \delta$. □

Starting the sequence at $e_0 = 0$ makes the first test fail if a single error falls in the first emitted set, and because the sequence stops at the first failure, that is an abort. An analytic start at $e_0 \geq 1$ removes this sensitivity to one error but places the first hypothesis at a more permissive threshold, which a weak score may fail for a different reason. Neither is better in general, which is why §6.4 reports the whole range.

4.4. Clustered units

When an image carries several questions, the item is not an i.i.d. unit; the image is. Two exact options exist. If calibration uses one item drawn uniformly at random from each of n i.i.d. images, the units are i.i.d. and Theorem 1 applies to the risk under that sampling scheme, which equals the item-level risk when images have equal numbers of items. If calibration uses all items, positive within-image correlation of errors makes $V_j \mid K_j$ over-dispersed relative to the binomial and the bound is anti-conservative by an amount that grows with the intra-image correlation; §6.10 measures this in a known-population simulation and §5 measures the correlation in the data we use.

Algorithm 1 CCRC. Folds $\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{cal}}$ are disjoint and unions of images; α, δ, e_0, c are fixed in advance.

fit s on $\mathcal{D}_{\text{fit}}$; compute out-of-fold scores $\tilde{s}$ on $\mathcal{D}_{\text{fit}}$
choose gate fraction q (external, fit-fold, or union); $g \leftarrow Q_{1-q}\{m_i : i \in \mathcal{D}_{\text{fit}}\}$
$\lambda_1 \leftarrow \min\{0.9, \max\{0.05,\, c\, k_{\text{start}}(e_0)/|\mathcal{D}_{\text{cal}}|\}\}$; $t_j \leftarrow Q_{1-\lambda_j}(\tilde{s})$, $j = 1..M$
$\hat{\jmath} \leftarrow 0$
for $j = 1$ **to** M **do**
    $A \leftarrow \{i \in \mathcal{D}_{\text{cal}} : s_i \geq t_j\}$, $R \leftarrow \{i \notin A : m_i \geq g\}$; $K \leftarrow |A| + |R|$
    $V \leftarrow \sum_A (1 - Y_i) + \sum_R (1 - Y_i^{(2)})$
    if $K > 0$ **and** $U(V, K; \delta) \leq \alpha$ **then** $\hat{\jmath} \leftarrow j$
    else break
    end if
end for
deploy: if $\hat{\jmath} = 0$ abstain everywhere; else accept if $s(x) \geq t_{\hat{\jmath}}$, repair if $m(x) \geq g$, abstain otherwise

4.5. The rank-indexed variant

An implementer who has no pre-specified grid takes thresholds as quantiles of the calibration sample itself, $t_j = Q_{1-\lambda_j}(s_{\text{cal}})$, and a repair gate $g = Q_{1-q}(m_{\text{cal}})$. Selecting the top $\lceil \lambda n \rceil$ items by score makes K deterministic and makes the emitted set a function of all calibration features jointly; then V is Poisson-binomial rather than binomial and the argument above does not apply. We treat this variant as the conventional baseline and quantify its behaviour in §6.2 and §6.10, without claiming a theorem for it.

5. Experimental protocol

Models, benchmarks and signals. Backbones are LLaVA-1.5-7B [83] in 4-bit, Qwen2-VL-2B-Instruct [84], and LLaVA decoded with visual contrastive decoding [25]. Channel 2 is OWLv2-base [18] with $\theta = 0.15$. Benchmarks are the POPE adversarial split [20] and the AMBER discriminative benchmark [12];

Table 2: Cohort flow. "Detector value" is the number of items for which the detector returned a score; the remaining items stay in every analysis, keep the channel-1 decision, and cannot be repaired. In POPE the exclusions are object phrases the extraction script failed to parse (POPE contains the typo "imange"); in AMBER they are the attribute and relation queries, which name no object. ICC is the intra-image correlation of the model's errors.

Setting	items	images	items/img	ICC	detector value	excluded from the grounded cohort	images/fold	μ (%)
POPE-1500, LLaVA	1500	250	6.0	−0.04	1500	0 unparsed object phrases	83/83/84	18.1
POPE-3000, LLaVA	3000	500	6.0	−0.02	3000	0 unparsed object phrases	166/166/168	17.0
POPE-adv, LLaVA	450	75	6.0	−0.05	444	6 unparsed object phrases	25/25/25	17.3
POPE-adv, Qwen2-VL	600	100	6.0	−0.04	591	9 unparsed object phrases	33/33/34	12.7
POPE-adv, LLaVA+VCD	600	100	6.0	−0.04	591	9 unparsed object phrases	33/33/34	19.7
AMBER(d), LLaVA	3000	957	3.1	−0.01	1311	1689 attribute/relation queries	319/319/319	23.6

MME, GQA and HallusionBench appear only where the detector does not apply (§6.7). Per-item signals were extracted once and are released, so every analysis below runs on a CPU. The extended POPE and AMBER cells (items 1500–2999 of the POPE adversarial split and items 500–2999 of the shuffled AMBER list) were extracted with the same prompts and detector query as the earlier cells; in the AMBER extension, photographs larger than 2000 pixels on the long side were shrunk before detection because the detector's preprocessing of the largest ones (54 megapixels) exhausted host memory.

Cohorts and the unit of splitting. The extraction streamed the first N rows of each public benchmark. We recovered the image identifier of every item from the benchmark metadata and verified the reconstruction against the cached gold labels, object phrases and categories (all agree). Table 2 reports what enters each cell. The POPE items are six questions per image (three present, three absent objects), so there are 75, 100, 250 and 500 distinct images in the four POPE cell sizes; the smaller POPE cells are subsets of the largest one's images, and POPE-3000 is the whole adversarial split. AMBER contributes 3.1 items per image over 957 images. Folds are unions of whole images, drawn at random with a third of the images per fold. The intra-image correlation of the model's errors is slightly *negative* (-0.01 to -0.09): in POPE the balanced yes/no design makes an error on one question of an image mildly predictive of a correct answer on another.

Primary analysis and missing evidence. The primary cohort contains all items. Items with no detector value receive zeroed grounding features and a missing-evidence flag in the score, and a margin $m = -1$ so they are never repaired. A secondary analysis restricted to items with a detector value appears in Appendix Appendix A.6. Excluding the items would analyse the detector's coverage; a deployed system must decide what to do with a question the detector cannot parse.

Procedure. Each split draws three image-disjoint folds (fit, calibration, test); the score is a logistic regression with five features (the VLM's probability, its confidence margin, the detector score, the detector score signed by the VLM's answer, and the missing flag); thresholds and gate are fixed from the fit fold as in §4.2; the fixed-sequence test uses Clopper–Pearson bounds at $\delta = 0.10$; and the deployed policy is evaluated on the test fold. We use 400 paired splits per cell, so every arm in a comparison sees identical folds.

Uncertainty. Repeated random splits of one benchmark measure how sensitive the procedure is to the split. They do not measure how the result would change on a fresh benchmark: the interval over splits shrinks like $1/\sqrt{\text{splits}}$ while uncertainty about the dataset does not shrink at all. We therefore report gains as the mean over the 400 splits with a 95% percentile interval from an *outer image-level bootstrap*: 300 bootstrap datasets are drawn by resampling images with replacement, the full protocol is run on each with 20 splits, and the interval is taken over bootstrap datasets. A duplicated image keeps its identifier and is always placed in one fold. Because a bootstrap dataset contains only about 63% distinct images, the interval is somewhat wider than the one a fresh sample of the same size would give. We do not report p-values over splits.

Validity audits. Realised test-fold risk carries binomial noise and cannot audit a population guarantee: a policy whose true risk is 0.08 exceeds 0.10 on a 150-item test fold with substantial probability. We report two statistics per cell. *exc* is the fraction of non-aborted splits whose test-fold risk exceeds α; under validity

it is bounded by δ plus finite-fold noise. *cv* is the fraction of splits for which the one-sided 90% Clopper–Pearson *lower* bound on the test-fold risk exceeds α (a confident violation); if the population risk is at most α it is at most 0.1 per split by the same argument as the theorem. Test folds are used for nothing else: no score, threshold, gate or start is selected on them. An exact audit requires a known population, which the simulation in §6.10 provides.

6. Results

6.1. Certified coverage

Table 3 compares CCRC with filtering on identical folds at $\alpha \in \{0.10, 0.15\}$, $\delta = 0.10$, all items, and Figure 3 plots the gains with their bootstrap intervals. Three certified gate variants appear. In the *external* variant the gate fraction q is a constant chosen once on a different benchmark (AMBER for the POPE cells, POPE-3000 for AMBER), by running the same protocol there on random subsamples whose calibration fold has the *target's* size and keeping the candidate (filtering, or $q \in \{0.05, 0.10, 0.25, 0.50\}$) with the highest mean certified coverage. The choice is $q = 0.05$ for the small cells at $\alpha = 0.10$, the small LLaVA cell at 0.15 and POPE-1500 at 0.10 ($q = 0.05$ in five of the twelve combinations) and $q = 0.5$ in the other seven. In the *fit-fold* variant q is chosen inside each split on the fit fold, with filtering as one candidate. A third variant (*external, full-size*) treats the whole development benchmark as a single calibration sample; it is reported in the last column of Table 3 because it shows what picking the gate at the wrong sample size costs. The size-matched variant was defined after we saw that the full-size variant loses on the small cells, so the comparison between them is exploratory and the size-matched rule should be read as a recommendation.

With the size-matched external gate the gain is positive in all twelve (cell, α) combinations, from +0.8 to +7.3 points, and the share of bootstrap datasets with a positive gain is 0.60–1.00. The 95% interval excludes zero in seven of the twelve: POPE-1500 and POPE-3000 at $\alpha = 0.15$, POPE-1500 at 0.10, AMBER

Table 3: Certified coverage (%), with the abort rate (%) in parentheses, $\delta = 0.10$, 400 paired splits, all-item cohort. Gain is the mean paired difference over splits against filtering; the interval is the 95% percentile interval from the outer image-level bootstrap (§5), and $P(\text{gain} > 0)$ is the fraction of bootstrap datasets with a positive gain. The last column is the gain with the gate chosen on the development benchmark at its full size.

		Filter	CCRC, external gate				CCRC, fit-fold gate		CCRC, full-size
Setting	α	cov. (abort)	cov. (abort)	gain [95% CI]	$P(\text{gain} > 0)$		cov. (abort)	gain [95% CI]	ext. gate gain
POPE-1500, LLaVA	0.10	68.2 (0)	69.6 (0)	+1.4 [+0.6, +7.3]	1.00	77.0 (1)	+8.8 [−8.0, +18.8]	+9.4	
	0.15	85.5 (0)	92.8 (0)	+7.3 [+4.2, +11.0]	1.00	92.6 (0)	+7.1 [+3.6, +10.7]	+7.3	
POPE-3000, LLaVA	0.10	76.6 (0)	83.2 (0)	+6.6 [−1.0, +10.7]	0.97	82.8 (0)	+6.3 [−2.3, +9.8]	+6.6	
	0.15	90.8 (0)	95.8 (0)	+5.1 [+2.9, +7.1]	1.00	95.8 (0)	+5.0 [+2.7, +7.0]	+5.1	
POPE-adv, LLaVA	0.10	44.5 (25)	46.2 (24)	+1.7 [+0.1, +5.8]	0.99	41.4 (34)	−3.1 [−13.8, +9.3]	−7.2	
	0.15	72.5 (2)	74.4 (1)	+1.8 [+0.1, +8.8]	0.99	82.0 (2)	+9.4 [−3.3, +28.8]	+10.1	
POPE-adv, Qwen2-VL	0.10	76.9 (1)	78.0 (0)	+1.2 [+0.1, +5.7]	1.00	71.2 (10)	−5.7 [−25.1, +10.5]	−7.7	
	0.15	94.6 (0)	95.7 (0)	+1.1 [−9.8, +13.2]	0.78	95.0 (0)	+0.4 [−7.5, +8.9]	+1.1	
POPE-adv, LLaVA+VCD	0.10	45.8 (14)	46.6 (12)	+0.8 [+0.0, +4.2]	0.90	39.6 (30)	−6.2 [−13.4, +8.9]	−12.3	
	0.15	74.4 (0)	76.8 (0)	+2.4 [−12.7, +22.2]	0.82	76.1 (0)	+1.7 [−10.3, +17.6]	+2.4	
AMBER(d), LLaVA	0.10	64.4 (0)	67.5 (0)	+3.1 [−14.5, +4.9]	0.60	66.8 (1)	+2.4 [−12.5, +4.4]	−5.5	
	0.15	75.7 (0)	79.6 (0)	+3.9 [+2.6, +8.3]	1.00	79.5 (0)	+3.8 [−0.4, +8.1]	+3.9	

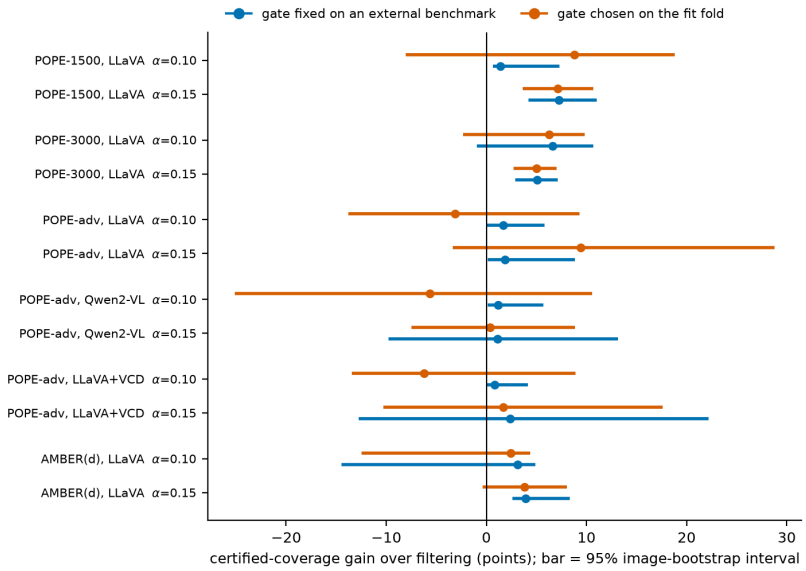


Figure 3: Gain in certified coverage over filtering. Points are means over 400 splits of the original benchmark; bars are 95% image-bootstrap intervals.

at 0.15, and three of the small cells, where the gain is +1.2 to +1.8 points with intervals whose lower end is within 0.15 points of zero. The larger gains occur where the calibration folds are large and the gate is $q = 0.5$: +6.6 and +5.1

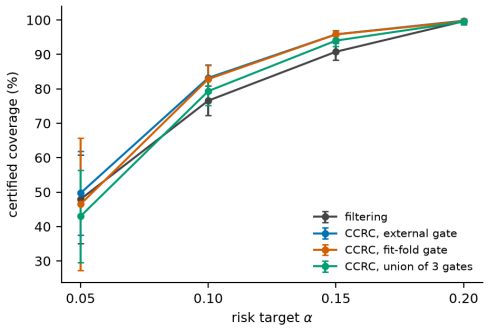


Figure 4: Certified coverage against the risk target on POPE-3000. Bars are one standard deviation over splits.

points on POPE-3000 (folds of about 1000 items, 166 images), +7.3 on POPE-1500 at $\alpha = 0.15$ and +3.9 on AMBER at $\alpha = 0.15$. On the small cells the selected gate is $q = 0.05$, which repairs 0.2–0.7% of test items and gains about one point; the per-split standard deviations of the gain are 1.3–11 points.

The fit-fold gate is more flexible and more variable: +8.8 and +7.1 points on POPE-1500 and +6.3, +5.0 on POPE-3000, but between −6.2 and +9.4 on the small cells, negative in all three at $\alpha = 0.10$, because a gate chosen on a fit fold of 150 items is itself noisy and a wrong choice costs more than the right one returns. The union over three gates avoids selection noise and pays $\delta/3$ per gate: it gains +2.0 to +4.4 points on the cells with 500 or more calibration items and loses up to 16.5 points on those with 150–200 (Appendix Appendix A.1). The full-size external gate ($q = 0.5$ for every POPE cell) loses 7.2, 7.7 and 12.3 points on the three small cells at $\alpha = 0.10$ and 5.5 on AMBER at 0.10, and gains up to +10.1 elsewhere. The continuity gate $q = 0.10$, selected in exploratory runs on these benchmarks, gains 0.0 to +3.9 points; we report it as an uncertified reference arm and do not use it for any claim, since a gate chosen with knowledge of the benchmarks is what Theorem 1 does not cover.

Coverage rises with α for every arm (Figure 4), and the gap between CCRC and filtering is largest at moderate targets. The audit columns of Appendix Table A.16 are discussed in §6.10.

Table 4: Filtering / CCRC certified coverage (%) at $\alpha = 0.10$ as the protocol changes. Rungs L0–L3 use the fixed gate $q = 0.10$, rung L4 chooses the gate on the fit fold, and rung L5 keeps items with no detector value. L0–L4 use the grounded cohort of each cell; L5 uses all items, so its absolute level for AMBER is lower: the attribute and relation queries are harder and cannot be repaired.

	POPE-1500, LLaVA	POPE-3000, LLaVA	POPE-adv, LLaVA	POPE-adv, Qwen2-VL	POPE-adv, LLaVA+VCD	AMBER(d), LLaVA
conventional protocol (item folds, calibration quantiles, e_0=0, fixed q=0.10)	68.6/71.8	77.2/79.2	44.0/47.1	77.9/79.7	48.8/51.0	91.0/26.2
+ image-grouped folds	68.2/71.3	77.2/79.2	43.1/48.3	77.6/79.5	47.9/50.8	90.8/34.7
+ fixed value grid and gate (fit fold)	67.5/70.9	76.5/78.5	35.0/42.3	75.5/77.7	39.9/45.3	88.3/78.3
+ analytic start e_0=1	68.2/71.1	76.6/78.6	43.6/47.4	75.6/78.0	45.5/47.6	90.9/90.1
+ gate chosen on the fit fold	68.2/77.0	76.6/82.8	43.6/39.4	75.6/68.3	45.5/39.7	90.9/93.3
+ all-item cohort	68.2/77.0	76.6/82.8	44.5/41.4	76.9/71.2	45.8/39.6	64.4/66.8

6.2. What each protocol change does

Table 4 introduces the changes one at a time, from a conventional implementation (item-level folds, thresholds and gate as quantiles of the calibration fold, a start at $k_{\text{start}}(0)$ and a fixed $q = 0.10$) to the protocol of this paper; the $\alpha = 0.15$ version is in Appendix Appendix A.2.

Image-grouped folds change coverage by at most 1.2 points on the POPE cells and lower the test-fold exceedance on the five POPE cells from $0.14, 0.10, 0.11, 0.16, 0.12$ to $0.10, 0.08, 0.09, 0.09, 0.12$. Replacing calibration quantiles by fit-fold values costs coverage when the fixed sequence starts at $e_0 = 0$: the first hypothesis lands at a threshold where a single error aborts the test, and the abort rate of filtering on POPE-adv LLaVA rises from 18% to 34%; the analytic start at $e_0 = 1$ recovers most of it. The gain of repair over filtering on the large POPE cells is stable under L0–L3 (+2.9 to +3.5 points on POPE-1500, +2.0 on POPE-3000). The largest effect of the protocol is on AMBER, where the conventional protocol makes CCRC abort on 72% of splits (coverage 26.2% against 91.0% for filtering); image-grouped folds, value-indexed thresholds and the analytic start remove it in turn (34.7%, 78.3%, 90.1%), and the fit-fold gate gives 93.3% against 90.9%. The loss that a conventional protocol attributes to repair on AMBER is therefore a property of the start of the fixed sequence.

6.3. What a repair is

A natural question for any correction method is how often it changes the model's answer, and whether the changes are right. Table 5 separates the re-

Table 5: Accounting of repaired items, external-gate variant, all items, pooled over test folds of 400 splits. "Repaired" and "answer changed" are percentages of test items; the next five columns are percentages of repaired items. "FP corrected" is the number of the model's false-positive errors (it said yes, gold is no) whose emitted answer was corrected, over the number of such errors; "FN corrected" is the share of the model's false-negative errors corrected.

			% of test items		% of repaired items						
Setting	α	q	repaired	answer changed	changed	wrong→right	right→wrong	agree, right	agree, wrong	FP corrected	FN corrected (%)
POPE-1500, LLaVA	0.10	0.05	0.86	0.45	52	52	0.0	48	0.0	0/11482	3.6
	0.15	0.5	6.42	4.21	65	57	8.2	32	2.5	579/11482	27.4
POPE-3000, LLaVA	0.10	0.5	10.08	5.72	57	46	11.0	40	3.0	1024/20516	36.3
	0.15	0.5	4.58	2.92	64	56	7.8	34	2.3	641/20516	20.1
POPE-adv, LLaVA	0.10	0.05	0.71	0.19	27	27	0.0	73	0.0	0/2960	1.8
	0.15	0.05	0.54	0.19	36	36	0.0	64	0.0	0/3828	1.8
POPE-adv, Qwen2-VL	0.10	0.05	0.43	0.00	0	0	0.0	100	0.0	0/2715	0.0
	0.15	0.5	1.43	0.47	33	26	7.1	53	13.6	0/2724	4.0
POPE-adv, LLaVA+VCD	0.10	0.05	0.20	0.16	78	78	0.0	22	0.0	0/9215	2.3
	0.15	0.5	6.96	2.80	40	30	10.3	52	7.8	864/10477	15.2
AMBER(d), LLaVA	0.10	0.5	4.37	2.38	54	51	3.6	35	10.5	1461/34791	12.5
	0.15	0.5	3.34	2.12	64	60	3.1	30	6.8	1467/34989	11.1

paired items (those emitted through channel 2) into those where channel 2's answer differs from the model's and those where the two agree, and counts wrong-to-right and right-to-wrong changes.

The accounting depends strongly on the gate. With the small gate ($q = 0.05$, selected on the cells with 150–200 calibration items at $\alpha = 0.10$) the repaired mass is 0.2–0.9% of test items and the answer is changed for 0.0–0.45%; on POPE-adv Qwen2-VL the detector agrees with the model on all repaired items and no answer changes. With the large gate ($q = 0.5$) the repaired mass is 4.4% (AMBER) to 10.1% (POPE-3000) of test items at $\alpha = 0.10$ and the answer changes for 2.4–5.7%. Among repaired items the detector agrees with the model on 22–100% at $\alpha = 0.10$: those items are *re-emitted with detector confirmation*, which is a different benefit from correction: an item the correctness score would have abstained on is emitted because a distinct source confirms it. Where the answer is changed it is mostly right: 18,583 wrong-to-right against 4,486 right-to-wrong changes on POPE-3000 (81%) and 8,830 against 634 on AMBER (93%), while with $q = 0.05$ the right-to-wrong count is zero in every cell. A larger gate corrects more and makes more mistakes, and the certificate prices the mixture. False-positive errors are corrected only with the large gate: 1,024

Table 6: Gate polarity at $\alpha = 0.10$, $q = 0.10$ per polarity. Gain over filtering in points, repaired items as a percentage of test items, and the number of false-positive errors corrected (pooled over 400 test folds). Pooled margin is the gate used elsewhere in the paper.

	pooled margin (CCRC)			positive only			negative only			symmetric		
Setting	gain	rep. %	FP fixed	gain	rep. %	FP fixed	gain	rep. %	FP fixed	gain	rep. %	FP fixed
POPE-1500, LLaVA	+2.9	1.88	0	+1.3	0.80	0	+0.2	0.07	0	+1.4	0.87	0
POPE-3000, LLaVA	+2.0	1.45	0	+0.6	0.43	0	+0.0	0.01	0	+0.6	0.44	0
POPE-adv, LLaVA	+3.7	1.93	0	+1.6	0.61	0	+0.8	0.14	0	+2.1	0.76	0
POPE-adv, Qwen2-VL	+2.6	0.90	0	+1.0	0.40	0	+2.5	0.20	0	+3.2	0.59	0
POPE-adv, LLaVA+VCD	+2.0	0.68	0	+0.8	0.19	0	+0.9	0.21	0	+1.5	0.40	0
AMBER(d), LLaVA	−0.0	3.24	0	+0.5	0.37	0	+0.0	0.00	8	+0.5	0.37	8

of 20,516 (5.0%) on POPE-3000 and 1,461 of 34,791 (4.2%) on AMBER, none with $q = 0.05$; 0 to 36.3% of the model's false-negative errors are corrected, the highest share on POPE-3000.

Table 6 repeats the accounting for one-sided and polarity-symmetric gates, each gating a fixed fraction $q = 0.10$ of its own polarity's margins. The symmetric gate admits negative-polarity repairs, which are highly precise, and these confirm the model's own "no" answers; they change an answer for 8 false positives on AMBER and none on the POPE cells. The reason is structural in this instantiation: the model's false positives are items where the detector also fires, so the detector's strongly negative evidence occurs on items the model already got right.

Where the gain comes from. Repaired items are themselves emitted, so whenever the certified threshold does not move the gain equals the repaired mass at filtering's own threshold; a dilution term appears when repairs lower the emitted risk and let the threshold open further. The threshold differs between the two arms on 3–34% of splits and the dilution term is 0.06 to 1.5 points; the automatic term is the larger one in ten of the twelve combinations (Appendix Table A.20).

6.4. The start of the sequence

Table 7 varies the number of errors the first hypothesis can tolerate, $e_0 \in \{0, 1, 2, 3\}$ (so $k_{\text{start}} = 22, 38, 52, 65$), and the placement multiplier $c \in \{1.0, 1.5\}$,

Table 7: Certified coverage (%) and abort rate (%) at $\alpha = 0.10$ as the start of the fixed sequence varies, filtering and CCRC with the fit-fold gate (Appendix Appendix A.2 for $\alpha = 0.05$ and 0.15).

			POPE-1500, LLaVA		POPE-3000, LLaVA		POPE-adv, LLaVA		POPE-adv, Qwen2-VL		POPE-adv, LLaVA+VCD		AMBER(d), LLaVA	
c	e_0	k_{start}	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC
1.0	0	22	65.8 (4)	76.5 (1)	76.5 (0)	82.8 (0)	22.2 (54)	33.8 (45)	72.6 (5)	64.6 (19)	22.9 (50)	29.9 (44)	64.4 (0)	67.1 (1)
1.0	1	38	68.3 (0)	76.6 (1)	76.5 (0)	82.8 (0)	41.2 (24)	40.7 (32)	76.5 (0)	68.2 (13)	42.8 (17)	39.1 (30)	64.4 (0)	67.1 (1)
1.0	2	52	68.4 (0)	76.9 (1)	76.6 (0)	82.9 (0)	44.4 (24)	41.8 (33)	76.8 (0)	71.5 (10)	45.2 (15)	39.0 (30)	64.4 (0)	67.1 (1)
1.0	3	65	68.2 (0)	77.2 (0)	76.5 (0)	82.8 (0)	44.0 (28)	39.3 (40)	76.0 (3)	69.5 (13)	45.3 (16)	39.5 (31)	64.4 (0)	67.1 (1)
1.5	0	22	67.5 (0)	76.8 (1)	76.5 (0)	82.8 (0)	36.9 (30)	38.4 (36)	76.5 (0)	68.6 (14)	39.5 (22)	37.9 (32)	64.4 (0)	67.1 (1)
1.5	1	38	68.2 (0)	77.0 (1)	76.6 (0)	82.8 (0)	44.5 (25)	41.4 (34)	76.9 (1)	71.2 (10)	45.8 (14)	39.6 (30)	64.4 (0)	66.8 (1)
1.5	2	52	68.4 (0)	77.1 (1)	76.5 (0)	82.9 (0)	38.5 (39)	36.2 (45)	77.1 (4)	70.0 (14)	43.8 (20)	37.6 (35)	64.4 (0)	66.7 (2)
1.5	3	65	68.3 (0)	77.5 (0)	76.6 (0)	82.9 (0)	22.4 (67)	22.2 (68)	78.6 (2)	73.7 (10)	33.0 (45)	29.4 (52)	64.3 (0)	67.1 (1)

for filtering and CCRC with the fit-fold gate. On the three cells with at least 250 images (POPE-1500, POPE-3000, AMBER) the choice is nearly irrelevant for CCRC (76.5–77.5%, 82.8–82.9% and 66.7–67.1%) and moves filtering by at most 2.6 points. On the three cells with calibration folds of 150–200 items filtering ranges from 22% to 45% (POPE-adv LLaVA), 23% to 46% (VCD) and 73% to 79% (Qwen2-VL), with abort rates up to 67%, and CCRC follows. Neither extreme is good: $e_0 = 0$ makes the first test fail on a single error; large e_0 together with $c = 1.5$ places the first hypothesis at a permissive threshold at which the score does not separate. We use $e_0 = 1$, $c = 1.5$ throughout, fixed before the final runs and following the analytic argument of Lemma 1; the table shows it lies in a reasonable region and it was not selected on test outcomes. The practical consequence is that conclusions drawn on calibration folds of 150–200 items depend on the start of the sequence more than on the repair.

A search that spends part of δ on several starts, instead of committing to one, is also valid (Corollary 2). We tested the union over $e_0 \in \{0, 1, 2\}$, each at $\delta/3$, choosing the certified policy with the largest calibration coverage (Appendix Table A.21). It is below the single analytic start $e_0 = 1$ in all twelve combinations, by 0.7 to 13 points (1–4 points on the cells with 500 or more calibration items, up to 13 on the small ones), because the $\delta/3$ penalty exceeds what the extra starts recover.

Table 8: POPE-1500, filtering only: certified coverage (%) with abort rate in parentheses, and area under the risk–coverage curve at $\alpha = 0.10$ (lower is better).

Score	α=0.05	α=0.10	α=0.15	AURC
image–text similarity only (ConfLVLM scorer)	0.0 (100)	0.5 (94)	3.4 (72)	0.1621
VLM confidence only	2.4 (93)	41.0 (18)	75.1 (5)	0.0768
learned, VLM signals	10.6 (61)	58.8 (1)	80.7 (0)	0.0678
+ image–text similarity	9.0 (67)	56.8 (2)	80.5 (0)	0.0691
+ detection grounding	21.8 (44)	68.2 (0)	85.5 (0)	0.0562
+ both	18.8 (46)	67.6 (0)	85.5 (0)	0.0577

6.5. Where the coverage comes from

The score. Table 8 compares scores under the same protocol, filtering only. The scorer of ConfLVLM [4], image–text similarity, certifies 0.5% here at $\alpha = 0.10$; VLM confidence alone certifies 41.0%; a learned combination of the VLM's own signals 58.8%; adding standardised image–text similarity does not help (56.8%); adding detection grounding gives 68.2% and the lowest area under the risk–coverage curve (0.0562 against 0.0678). Against the similarity scorer, CCRC with the external gate (69.6%) gains 69.1 points, of which 67.7 come from the score and 1.4 from certified repair; with the fit-fold gate (77.0%) the repair share is 8.8 points. The correctness score is the dominant lever.

The detector as a predictor. If channel 2 is good, why not use it alone? Table 9 reports selective prediction on the detector alone. Its raw accuracy matches the VLM's on POPE (83.1% against 81.9%), but its selective coverage is lower (at $\alpha = 0.10$, 57.7 against 68.2% on POPE-1500, 68.0 against 76.6% on POPE-3000, 39.9 against 64.4% on AMBER, and 12–15% against 45–77% on the small cells): accuracy does not imply a separable score.

Composition with a mitigation decoder. On POPE-adversarial with visual contrastive decoding [25] (base risk 19.7%, the highest of our cells), CCRC with the external gate gains +0.8 and +2.4 points over filtering at $\alpha = 0.10$ and

Table 9: Certified coverage (%) of VLM filtering, CCRC (external gate) and detector-only selective prediction, and the detector's accuracy on grounded items.

	$\alpha = 0.10$			$\alpha = 0.15$			
Setting	VLM filter	CCRC	detector only	VLM filter	CCRC	detector only	detector acc.
POPE-1500, LLaVA	68.2	69.6	57.7	85.5	92.8	84.8	83.1
POPE-3000, LLaVA	76.6	83.2	68.0	90.8	95.8	88.6	83.2
POPE-adv, LLaVA	44.5	46.2	11.6	72.5	74.4	43.0	82.4
POPE-adv, Qwen2-VL	76.9	78.0	14.6	94.6	95.7	57.7	82.9
POPE-adv, LLaVA+VCD	45.8	46.6	14.6	74.4	76.8	57.7	82.9
AMBER(d), LLaVA	64.4	67.5	39.9	75.7	79.6	41.2	92.3

0.15, with the sign supported by 90% and 82% of bootstrap datasets (Table 3). Lowering the base risk and adding certified repair are compatible; neither result is large.

6.6. Self-certified repair

A design that needs no second channel would flip the model's answer where the score says it is probably wrong. For binary answers the flipped answer is wrong exactly when the model was right, so the flip region's risk is the model's *accuracy* there.

Proposition 2 (Self-repair is bounded by slack)**.** *Let the repair be the negation of the model's answer on a region R of mass c_r and let the accepted region have mass c_a and risk r_a. The risk of the repaired region is $r_r = \mathbb{P}[Y = 1 \mid X \in R]$. If $r_r > \alpha$, the emitted mixture has risk at most α only if $c_r \le c_a(\alpha - r_a)/(r_r - \alpha)$.*

Proof. On R the emitted answer is $1 - \hat{a}$, correct iff $\hat{a}$ is wrong, so its error indicator is Y and averaging gives r_r. The mixture risk $(c_a r_a + c_r r_r)/(c_a + c_r) \le \alpha$ rearranges to $c_r(r_r - \alpha) \le c_a(\alpha - r_a)$. □

Table 10 flips the answer on the bottom $q \in \{0.02, 0.05, 0.10\}$ of the score, at both starts. The flip region's risk is 12–56%, far above α, on every cell except AMBER, and certified coverage falls by 4 to 77 points at $\alpha = 0.10$ with the analytic start, with abort rates up to 100%. On AMBER the bottom 2% of the score is clean (flip-region risk 2%) and self-repair gains +3.1 points there,

Table 10: Self-certified repair against filtering at $\alpha = 0.10$: gain in points (abort rate, %) and the risk of the flip region r_{flip} (%), at both starts; the last column is CCRC with the $q = 0.10$ detector gate.

			q=0.02		q=0.05		q=0.10		
Setting	e_0	filter (ab)	gain (ab)	r_{flip}	gain (ab)	r_{flip}	gain (ab)	r_{flip}	CCRC gain
POPE-1500, LLaVA	0	67.5 (0)	−53.2 (79)	31	−66.9 (99)	36	−67.5 (100)	–	+3.5
	1	68.2 (0)	−35.1 (51)	29	−64.0 (94)	37	−68.0 (100)	43	+2.9
POPE-3000, LLaVA	0	76.5 (0)	−56.9 (75)	25	−76.5 (100)	–	−76.5 (100)	–	+2.0
	1	76.6 (0)	−50.7 (67)	26	−76.6 (100)	–	−76.6 (100)	–	+2.0
POPE-adv, LLaVA	0	36.9 (30)	−28.6 (84)	54	−34.8 (96)	51	−36.9 (100)	57	+7.1
	1	44.5 (25)	−25.6 (68)	50	−39.2 (91)	50	−44.1 (99)	50	+3.7
POPE-adv, Qwen2-VL	0	76.5 (0)	−60.0 (75)	56	−72.7 (94)	53	−76.0 (99)	61	+2.7
	1	76.9 (1)	−51.4 (63)	55	−70.3 (90)	53	−76.3 (99)	56	+2.6
POPE-adv, LLaVA+VCD	0	39.5 (22)	−6.8 (39)	12	−27.6 (78)	25	−38.8 (98)	35	+4.6
	1	45.8 (14)	−4.1 (26)	12	−25.7 (64)	26	−43.2 (96)	39	+2.0
AMBER(d), LLaVA	0	64.4 (0)	+3.1 (0)	2	−10.4 (22)	12	−64.4 (100)	–	−1.4
	1	64.4 (0)	+3.1 (0)	2	−8.6 (19)	13	−64.4 (100)	–	−0.0

while the bottom 5% (13%) already loses 8.6. The fixed-mass, high-error block is largest relative to the emitted set at the beginning of the sequence, fails the first hypothesis, and the sequence cannot continue to the later thresholds where Proposition 2 would allow it. The start does not change the conclusion: at $e_0 = 0$ the losses are 7–77 points. Self-repair is not impossible in principle (the proposition admits it when the accepted region has slack and the flip region is clean), but on all cells except AMBER no such region exists at the target of 0.10.

6.7. Items without usable evidence

Table 2 shows that 0 of 3000 POPE-3000 items but 1689 of 3000 AMBER items have no detector value. Where the detector never applies, CCRC reduces to filtering. Appendix Table A.22 gives filtering on three benchmarks where the detector returns nothing (MME, GQA, HallusionBench): at $\alpha = 0.10$ coverage is 0.9%, 8.5% and 0% with abort rates of 97%, 72% and 100%. On HallusionBench the base risk is 48.8% and nothing is certifiable at $\alpha \leq 0.20$; the floor alone is $(0.488 - 0.20)/0.80 = 36\%$ and the abstention is far larger. Applicability of the evidence channel depends on the benchmark's question types.



Figure 5: Accepted answers that are wrong under the policy of Figure 1: two confident false positives and two confident misses (LLaVA-1.5 on POPE, held-out images).

6.8. Examples and failures

Figure 1 shows accepted, repaired and abstained items. The cases that matter for the guarantee are the accepted answers that are wrong, which the certificate allows at rate α among emitted items (Figure 5). Two kinds recur. A confident false positive (the model says the object is present with high probability and the score is high) is accepted whether the detector disagrees (conf 0.14 for the car) or is fooled as well (conf 0.30 for a bus that is absent), because an absent object's margin stays below the repair gate. A confident miss on which the detector is also weak is accepted too: the model answers "no" with high confidence about an object that is present and the detector's confidence is low (here 0.03–0.10), so neither the score nor the evidence margin flags it. Both are the items that remain after any filtering rule; repair addresses only the low-score items on which the detector is decisive.

6.9. Channel overlap and a held-out benchmark

Channel 1 without detector features. In our instantiation the detector score is also an input to the correctness score, so the two channels are not independent. To see whether the repair gain depends on this overlap, Table 11 removes every detector feature from channel 1 (the score uses only the VLM's probability and confidence) and keeps the detector as channel 2. Filtering then certifies less (for example 58.8% against 68.2% on POPE-1500 at $\alpha = 0.10$), and CCRC with a

Table 11: Certified coverage (%, abort rate in parentheses) with channel 1 restricted to the VLM's own signals and with the full score. CCRC uses the size-matched external gate in both cases (selected with the matching score); gain is against filtering with the same score.

		channel 1 without detector features			channel 1 with detector features		
Setting	α	filter	CCRC	gain	filter	CCRC	gain
POPE-1500, LLaVA	0.10	58.8 (1)	72.0 (1)	+13.3	68.2 (0)	69.6 (0)	+1.4
	0.15	80.7 (0)	88.1 (0)	+7.5	85.5 (0)	92.8 (0)	+7.3
POPE-3000, LLaVA	0.10	67.7 (0)	77.4 (0)	+9.7	76.6 (0)	83.2 (0)	+6.6
	0.15	88.2 (0)	92.6 (0)	+4.5	90.8 (0)	95.8 (0)	+5.1
POPE-adv, LLaVA	0.10	28.5 (51)	31.4 (47)	+2.9	44.5 (25)	46.2 (24)	+1.7
	0.15	62.9 (11)	81.3 (2)	+18.4	72.5 (2)	74.4 (1)	+1.8
POPE-adv, Qwen2-VL	0.10	75.3 (0)	76.8 (0)	+1.5	76.9 (1)	78.0 (0)	+1.2
	0.15	95.3 (0)	95.6 (0)	+0.3	94.6 (0)	95.7 (0)	+1.1
POPE-adv, LLaVA+VCD	0.10	6.5 (88)	11.5 (78)	+5.0	45.8 (14)	46.6 (12)	+0.8
	0.15	25.2 (52)	71.4 (6)	+46.2	74.4 (0)	76.8 (0)	+2.4
AMBER(d), LLaVA	0.10	52.3 (0)	58.8 (0)	+6.5	64.4 (0)	67.5 (0)	+3.1
	0.15	66.0 (0)	72.3 (0)	+6.3	75.7 (0)	79.6 (0)	+3.9

size-matched external gate (chosen with the same detector-free score) gains +0.3 to +46 points over that filter, positive in all twelve combinations; the gain is larger than in the main table because the baseline is weaker. In absolute terms the detector-free CCRC certifies 72.0% against 69.6% for the full external-gate pipeline on POPE-1500 at 0.10 and 77.4% against 83.2% on POPE-3000, and is below the full pipeline in 10 of the 12 combinations by 0.1 to 35 points: using the detector for repair alone recovers much but not all of what using it in the score as well gives. The validity statistics stay in range (exceedance 0.05–0.20, except 0.39 for VCD at 0.10, where 88% of splits abort and the statistic rests on few splits; confident violations ≤ 0.03 except 0.10 there).

Score, grid and gate from another benchmark. Intervals over resampled images do not speak to new data. Table 12 therefore fits the correctness model, the score grid and the gate on one benchmark and uses the target only for calibration and test: AMBER as source for the POPE cells and POPE-3000 as source for AMBER. The guarantee covers this setting, since the policy family is

Table 12: Held-out benchmark. Score model, score grid and gate are fitted on the source benchmark; calibration and test folds come from the target. In-domain CCRC is the size-matched external-gate result of Table 3. exc: test-fold exceedance of α; cv: confident-violation rate.

Target ← source	α	in-domain CCRC	filter (abort)	CCRC (abort)	exc / cv	q
AMBER(d), LLaVA ← POPE-3000	0.10	67.5	0.0 (100)	48.5 (0)	0.12 / 0.02	0.5
AMBER(d), LLaVA ← POPE-3000	0.15	79.6	0.0 (100)	59.8 (0)	0.03 / 0.00	0.5
POPE-3000, LLaVA ← AMBER(d)	0.10	83.2	29.2 (11)	67.0 (0)	0.06 / 0.00	0.5
POPE-3000, LLaVA ← AMBER(d)	0.15	95.8	60.6 (0)	80.1 (0)	0.08 / 0.01	0.5
POPE-adv, LLaVA ← AMBER(d)	0.10	46.2	0.8 (98)	1.4 (98)	1.00 / 0.40	0.05
POPE-adv, LLaVA ← AMBER(d)	0.15	74.4	32.3 (22)	43.2 (6)	0.08 / 0.00	0.05
POPE-adv, Qwen2-VL ← AMBER(d)	0.10	78.0	16.0 (66)	26.9 (46)	0.13 / 0.01	0.05
POPE-adv, Qwen2-VL ← AMBER(d)	0.15	95.7	60.3 (0)	82.5 (0)	0.07 / 0.00	0.5
POPE-adv, LLaVA+VCD ← AMBER(d)	0.10	46.6	7.9 (83)	11.7 (75)	0.48 / 0.03	0.05
POPE-adv, LLaVA+VCD ← AMBER(d)	0.15	76.8	51.0 (5)	74.9 (0)	0.10 / 0.01	0.5

fixed from independent data. Test-fold exceedance is 0.03–0.13 in eight of the ten combinations and the confident-violation rate is at most 0.03 in nine; the exceptions are the two smallest POPE cells at $\alpha = 0.10$, where 75–98% of splits abort and the statistics rest on a handful of splits (for POPE-adv LLaVA the 2% of splits that do not abort are too few to audit, and the confident-violation rate there is 0.40). Filtering with the transferred score fails more often than in-domain, because the source benchmark's score is miscalibrated on the target, and the repair region (a gate set on detector margins, which transfer better) restores certifiability: CCRC certifies 48.5% and 59.8% on AMBER where transferred filtering certifies nothing. These coverage differences measure how badly a transferred score calibrates and should not be read as the gain of repair; the result of interest is that the certificate holds when the source of the score and gate is a different benchmark.

6.10. Auditing validity

Real data. Across all arms and cells in Appendix Table A.16 the test-fold exceedance of α is 0.03–0.23 and the confident-violation rate is at most 0.04, against the bound of 0.10 per split. We treat this as consistent with validity and do not treat it as proof: with test folds of 150–1000 items and risks near

Table 13: Known-population exceedance $\mathbb{P}[\mathcal{R} > \alpha]$ (aborts counted as compliant), $\alpha = 0.10$; target $\delta = 0.10$. "Value" fixes thresholds and gate from an independent fit sample (the object of Theorem 1); "rank" takes them as quantiles of the calibration sample; the last column calibrates on one random item per image.

ICC	μ	n_{cal}	value, filter	value, repair	rank, filter	rank, repair	value, repair, 1 item/image
0 (i.i.d.)	0.15	150	0.052	0.063	0.049	0.064	–
0 (i.i.d.)	0.15	450	0.063	0.051	0.051	0.058	–
0 (i.i.d.)	0.25	150	0.046	0.052	0.036	0.046	–
0 (i.i.d.)	0.25	450	0.047	0.050	0.058	0.055	–
0.08	0.15	150	0.060	0.071	0.063	0.076	0.053
0.08	0.15	450	0.065	0.078	0.090	0.067	0.051
0.08	0.25	150	0.050	0.068	0.053	0.067	0.045
0.08	0.25	450	0.054	0.067	0.056	0.078	0.052
0.26	0.15	150	0.078	0.102	0.069	0.103	0.041
0.26	0.15	450	0.068	0.111	0.081	0.110	0.056
0.26	0.25	150	0.027	0.080	0.031	0.099	0.042
0.26	0.25	450	0.043	0.085	0.043	0.081	0.048

α, exceedance of the realised fold risk is expected, and a real-data audit cannot exceed the resolution of its test fold.

Known population. The exact audit uses a generative population in which the risk of the selected policy is computed on 400,000 reference items. Items come in image clusters of six with a shared random effect that sets the intra-image correlation of errors (ICC); the score is the best item-level predictor; channel 2 is right with probability $1 - \frac{1}{2}e^{-4v}$ at margin v. We audit the exceedance $\mathbb{P}[\mathcal{R}(\pi_j) > \alpha]$, counting aborts as compliant, over 3000 trials per cell, $\mu \in \{0.15, 0.25\}$, $\alpha \in \{0.10, 0.15\}$, $n_{\text{cal}} \in \{150, 450\}$.

With i.i.d. items (ICC $= 0$) the exceedance is at most 0.064 in every cell for both the value-indexed family, which Theorem 1 covers, and the rank-indexed family, which it does not, so the rank-indexed variant is also conservative here: the missing proof for it is a gap in the argument rather than an observed failure. With positive within-image correlation the guarantee degrades: at ICC 0.08 the worst cell is 0.091, and at ICC 0.26 the exceedance of the value-indexed family rises to 0.125 (filtering) and 0.132 (repair), above δ, and the rank-indexed variant

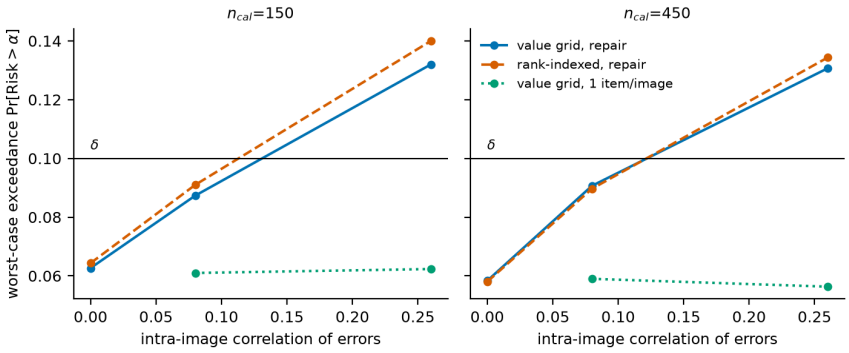


Figure 6: Worst-case known-population exceedance over the μ and α cells, against the intra-image correlation of errors, with repair. Horizontal line: $\delta = 0.10$.

to 0.14. Calibrating on one random item per image restores validity in every cell (≤ 0.062) at the price of a smaller calibration sample. In the benchmarks used here the measured ICC is slightly negative (Table 2), so these data are not in the clustered regime. A benchmark with several correlated questions per image would be, and there the item-level bound should not be used without the one-item-per-image calibration or a design-effect correction.

7. Comparison with concurrent methods

Two methods published in 2026 are close enough to CCRC that a reader will ask what separates them. Table 14 compares the three on the dimensions the reviewing literature cares about. The comparison is conceptual: the benchmarks, backbones and sample sizes differ, so we do not compare numbers across papers and we did not rerun either method.

BCEA. BCEA reduces the abstention floor's practical bite by spending a visual budget to re-read the image and recalibrating on post-acquisition scores. It stays inside the premise of Proposition 1: acquisition changes how confident the method is about the model's claim, and which claim is emitted does not change. Its current version certifies a Bonferroni-corrected Clopper–Pearson test over a threshold grid fixed independently of the calibration sample and reports certified

Table 14: CCRC, BCEA [5] (version 4, 21 July 2026) and CEBC [77].

	CCRC	BCEA	CEBC
Action on a flagged claim	emit a detector answer, accept, or abstain	acquire more visual evidence, then answer or abstain; the answer is the model's own	rewrite with a VLM, possibly substituting or inserting objects, then delete what remains below threshold
Quantity controlled	error rate of the emitted set, accepted and substituted answers together	error rate of emitted claims	per-mention false-accept rate of a detector threshold on hallucinated mentions in base captions
Guarantee	finite-sample, probability $1-\delta$ over calibration, i.i.d. units	finite-sample, Bonferroni-corrected Clopper–Pearson over a fixed grid; a full-level scan is labelled uncertified	none stated; no theorem, no sampling model
Revised output inside the guarantee	yes	not applicable (no revision)	no; the rewrite may use objects at $\tau - 0.05$ and is outside the calibration
Calibration procedure	fixed-sequence test over values from the fit fold	fixed grid, Bonferroni	one clipped quantile
Role of the detector	channel 2: supplies the substituted answer and its margin	optional guide for acquisition	sole arbiter of support

coverage of 0.23 without acquisition and 0.32 with gated acquisition at $\alpha = 0.10$; the coverage figures of 0.28 and 0.37 correspond to its uncertified practical variant, which scans the calibration scores at full confidence. The mechanisms are different (re-scoring a claim versus replacing it) and in principle composable, since a re-scored claim can still be repaired when the detector is decisive. We have not tested the composition in this version, and nothing here shows that it helps.

CEBC. CEBC is the nearest published method that edits unsupported object mentions in VLM output with an external detector. Its calibration is a one-sided quantile of detector scores on hallucinated mentions from base captions; its rewrite stage is not calibrated; and it reports no violation check. Whatever its empirical merit, it is a detector-gated rewriter rather than a certified selective predictor, and its headline numbers (for example CHAIR_S from 10.0 to 2.0) measure a different object than a certified error rate over emitted claims. CEBC applies to free-form captions and CCRC as evaluated to yes/no questions; the guarantee of Theorem 1 has no free-form counterpart in this paper, and §8 gives the reason.

What remains of the novelty claim. Among the work we surveyed up to 4 October 2026, CCRC is the one whose certificate covers an emitted answer that differs from the model's, with the repair gate fixed from fit or external data. CEBC edits without certifying; BCEA certifies without editing; conformal linguistic calibration [60] certifies a rewritten output for text-only question answering by generalising claims. The claim is about the combination, in the setting of object-existence questions, and not about the idea of correcting hallucinations.

8. Free-form generation: a matched-prefix test of mid-sequence repair

Correction looks more attractive in captions than in yes/no answers: replacing a hallucinated word keeps a caption that deletion would damage. The extension seems mechanical, and two obstacles are known. Editing a prefix makes later tokens off-policy, which is feedback covariate shift [78] and falls outside weighted conformal methods [79]; and per-claim guarantees say nothing about a whole sequence whose length is data-dependent. Prefix-conditional (Mondrian) p-values and e-processes [44, 85] could handle both in principle, leaving the drift of the calibration pool, which on-policy calibration by fixed-point iteration would address if the correction map were monotone. We measured whether it is before attempting a proof.

Table 15: Paired matched-prefix test ($n = 121$). All p-values are two-sided.

Downstream measure (continuation only)	Original	Repaired	paired difference (p)
Hallucinated mentions (mean)	1.207	1.413	+0.207 (0.025)
Hallucinations per 100 words	3.61	4.30	+0.75 (0.011)
Faithful mentions (mean)	1.579	1.752	+0.174 (0.077)
Continuation length (words)	33.4	32.9	−0.55 (0.092)
Hallucinated share of mentions	0.433	0.447	−0.004 (0.884)

Design. For each image we generate a caption greedily, locate the first hallucinated object mention at position t using AMBER's per-image present and absent object lists, and build two prefixes identical up to t: one that keeps the hallucinated word and one that substitutes the best-detected object the image contains. We continue decoding the same number of tokens from each and count hallucinated mentions in the continuations only. The only difference between the two continuations is the intervention. The unit is the matched pair ($n = 121$ images); there is one repair per pair.

Result. In this sample repair makes the continuation worse (Table 15, Figure 7). Hallucinated mentions in the continuation rise from 1.207 to 1.413, a paired difference of $+0.207 \pm 0.091$ ($t(120) = 2.27$, two-sided $p = 0.025$), with 45 pairs worsening, 25 improving and 51 tied. A Wilcoxon signed-rank test ($p = 0.035$) and an exact sign test ($p = 0.023$) agree. The Hodges–Lehmann estimate is 0.000 because 42% of pairs are tied.

What the experiment can support. The result is significant under three tests and under-powered. At the observed paired standard deviation (0.999) the minimum detectable effect at 80% power is +0.257, larger than the observed +0.207, and about 186 pairs would be needed at this effect size. Equivalence testing against a margin of ± 0.25 does not establish equivalence ($p = 0.317$). The sign is established and the magnitude is not: any value in $[+0.027, +0.386]$ is compatible. The number of hallucinated mentions rises together with faithful

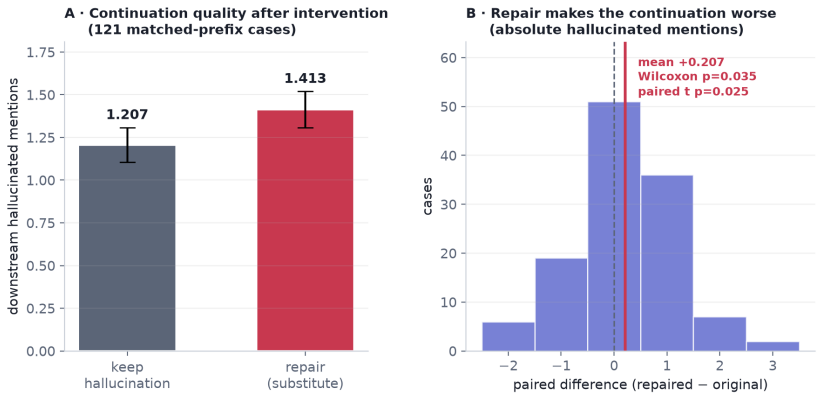


Figure 7: (A) Mean hallucinated mentions in the continuation after keeping or repairing the flagged claim (standard errors). (B) Distribution of the paired difference.

mentions and with mention density; per mention, the hallucinated share does not move ($0.433 \to 0.447$, $p = 0.884$), so the mechanism is denser object talk after repair. A certificate on the count of false claims is not helped by this, since each additional hallucinated mention counts. The substitution used here is context-blind, and the finding concerns that rule rather than all correction. What the data withdraw is support for a monotone correction map and hence for a fixed-point route to a sequential guarantee, at the sample size we have.

A coupling bound. For a sequence of claims let $F(O)$ be the number of false claims and $\mathcal{R}(O) = \mathbb{E}[F(O)]$ the expected count per sequence. Let a repair rule replace claims at a random set P of positions, producing O'. Write W for the number of wrong repaired claims, $F_P(O)$ for the number of false *original* claims at the repaired positions, and Δ for the signed change in the number of false claims at non-repaired positions between O' and O.

Theorem 2 (Coupling bound for corrected sequences)**.** $F(O') = F(O) - F_P(O) + W + \Delta$ *pointwise, hence*

$$\mathcal{R}(O') = \mathcal{R}(O) + \mathbb{E}[W] - \mathbb{E}[F_P(O)] + \mathcal{I} \;\le\; \mathcal{R}(O) + \mathbb{E}[W] + \mathcal{I}, \qquad \mathcal{I} := \mathbb{E}[\Delta].$$

$\mathcal{I}$ is signed: a repair that prevents later errors makes it negative. If at most one claim is repaired per sequence, $\mathbb{E}[W] = \mathbb{P}[X \in R]\,\mathbb{P}[\text{repair wrong} \mid X \in R]$; in

general $\mathbb{E}[W] = \sum_t \mathbb{P}[t \in P,\ \textit{repair at } t \textit{ wrong}]$ *is an expected count over repair opportunities.*

Proof. The false claims of O' are those at repaired positions (the wrong repairs, W) plus those at non-repaired positions. The false claims of O are those at repaired positions ($F_P(O)$) plus those at non-repaired positions. Subtracting gives the identity, and taking expectations gives the equality; $F_P(O) \geq 0$ gives the bound. □

In the matched-prefix experiment one claim is repaired per pair, so $\widehat{\mathcal{I}}$ is the paired difference in the continuation: $+0.207 \pm 0.091$ additional false claims per repair. That is as large as the direct term for a repair that is wrong with probability 0.2, and it cannot be dropped. Certifying $\mathcal{R}(O) \leq \alpha_1$ and the direct term by α_2 at $\delta/2$ each gives $\mathcal{R}(O') \leq \alpha_1 + \alpha_2 + \mathcal{I}$ at $1 - \delta$, which is useful only relative to an already-selective base policy, since certifying the uncorrected generation is the object the abstention floor makes expensive. A sequential procedure would have to price the continuation and not only the claim, for example by searching over rewrites under an accumulated risk ledger. We leave that open.

9. Discussion

What the guarantee says. With probability $1 - \delta$ over the calibration sample, the population risk of the deployed policy, accepted and substituted answers together, is at most α. It says nothing about a single item, about a finite test fold, about a distribution other than the one the units were drawn from, or about the coverage that results. A sampling model is part of it: units are i.i.d., and when a benchmark is split at random the iid assumption is a modelling choice about the population the benchmark represents.

What the evidence supports. Under a protocol in which the proved procedure is the evaluated procedure, certified repair gains between 0.8 and 7.3 points of coverage over certified filtering when the gate is fixed on external data at the

target's calibration size; the sign is supported by 60–100% of image-bootstrap datasets and the interval excludes zero in seven of twelve combinations. The gain is about one point when calibration folds hold 150–200 items and up to 7.3 points with 500 or more; where the interval includes zero, more images would be needed to size it. Repair does add coverage; a large share of it comes from confirming answers the model already gave, and false positives are corrected only with the large gate and then for about 4–5% of them. The correctness score dominates the coverage, and the start of the fixed sequence dominates the repair on small folds.

Scope, and other hallucination types. The method as evaluated repairs object-existence answers, and in practice it repairs missed objects; the title says so. Extending it needs a distinct evidence source for each claim type, with a margin that can gate repair. For attributes this could be a region-level attribute classifier or a second VLM queried on a crop; for counts, a detector's instance count compared against the model's; for relations, a scene-graph model. The certificate does not change, since it only needs a fixed policy and i.i.d. units, but Section 6.7 shows what happens without such a source: on attribute and relation questions CCRC reduces to filtering. Whether a source with a decisive margin exists for those claim types is an empirical question this paper does not answer.

Limitations. (i) The task is binary object existence, with one detector, three backbones and two benchmark families whose images overlap; 1,457 distinct images are involved in total. (ii) The channel-2 operating point $\theta = 0.15$ was fixed in the extraction scripts and is not varied here; we do not claim it was chosen independently of the benchmarks. (iii) The external gate for the POPE cells was chosen on AMBER, whose image sources we could not verify to be disjoint from POPE's COCO images, and the size-matched rule was defined after the full-size rule had been seen to fail on small cells; the guarantee requires the gate to be independent of the calibration sample, and any image overlap would weaken that independence slightly. (iv) The i.i.d. assumption is violated within images

when errors are correlated; here the measured correlation is slightly negative, and §6.10 shows the guarantee degrades with positive correlation unless calibration uses one item per image. (v) The bootstrap interval reflects sampling of images from the benchmark, not of benchmarks from the world. (vi) Free-form generation is not covered and mid-sequence repair worsens the continuation. (vii) We did not run BCEA or CEBC, we tested a detector-free channel 1 (Table 11) but not a second detector, and the signals of the two extended cells were extracted once, with images above 2000 pixels on the long side shrunk before detection in the AMBER extension.

Next steps. The measurement that would most improve the paper is a second detector and more benchmarks with several questions per image, which would narrow the intervals the bootstrap shows to be wide on the small cells. A repair rule that respects the generation context, tested on the free-form setting with the matched-prefix design, is the natural extension.

10. Conclusion

Certified correction replaces an answer the model would have abstained on with a detector's answer and certifies the emitted set as a whole; the procedure is valid under i.i.d. sampling when thresholds and gate are fixed before calibration. Under a protocol built on that requirement the coverage gain over filtering is positive and moderate (+0.8 to +7.3 points), with an uncertainty that the available images do not resolve on the smaller cells. Many repaired items re-emit the model's answer with detector confirmation, false positives are corrected only with a large gate and then for a small share of them, and the coverage comes mainly from the correctness score. In a small matched-prefix experiment, mid-sequence repair worsened the continuation. The reproducible pipeline and all negative results are released.

Data and code availability

Per-item model and detector signals, image identifiers and every script that produces the tables and figures are released at `https://github.com/vinhqdang/cmvkg` (directory `conformal_followup/revision`).

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used generative AI tools to support writing and coding. All ideas, algorithms and research directions are the author's own. The author reviewed and edited all content produced with these tools and takes full responsibility for the content of the publication.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] S. Bates, A. N. Angelopoulos, L. Lei, J. Malik, M. I. Jordan, Distribution-free, risk-controlling prediction sets, Journal of the ACM 68 (6) (2021) 43:1–43:34.

[2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, The Annals of Applied Statistics 19 (2) (2025) 1641–1662. doi:10.1214/24-AOAS1998.

[3] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal risk control, in: International Conference on Learning Representations (ICLR), 2024.

[4] Z. Li, C. Yan, N. J. Jackson, W. Cui, B. Li, J. Zhang, B. A. Malin, Towards statistical factuality guarantee for large vision–language models, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Suzhou, China, 2025.
URL https://aclanthology.org/2025.emnlp-main.576/

[5] J. Xu, Y. Wu, D. Zeng, J. Paisley, Q. Zhao, Look again before you abstain: Budgeted conformal evidence acquisition for reliable vision–language models, arXiv preprint arXiv:2606.16667Version 4, 21 July 2026 (2026).
URL https://arxiv.org/abs/2606.16667

[6] M. Atabuzzaman, C. Alexander, C. Thomas, IntroConformal: Conformal factuality guarantees for large vision-language models via introspective signals, arXiv preprint arXiv:2609.01375ArXiv comment: EMNLP 2026 main conference (proceedings entry not checked) (2026).
URL https://arxiv.org/abs/2609.01375

[7] S. Yin, C. Fu, S. Zhao, T. Xu, H. Wang, D. Sui, Y. Shen, K. Li, X. Sun, E. Chen, Woodpecker: Hallucination correction for multimodal large language models, Science China Information Sciences 67 (12) (2024) 220105:1–

220105:13.
URL https://doi.org/10.1007/s11432-024-4251-x

[8] Y. Zhou, C. Cui, J. Yoon, L. Zhang, Z. Deng, C. Finn, M. Bansal, H. Yao, Analyzing and mitigating object hallucination in large vision-language models, in: International Conference on Learning Representations (ICLR), Vienna, Austria, 2024.

[9] T.-H. Wu, H. Lee, J. Ge, J. E. Gonzalez, T. Darrell, D. M. Chan, Generate, but verify: Reducing hallucination in vision-language models with retrospective resampling, in: Advances in Neural Information Processing Systems (NeurIPS), 2025.
URL https://reverse-vlm.github.io/

[10] Q.-V. Dang, N.-S.-A. Nguyen, T.-B.-D. Vo, M. N. Dinh, A. D. Le, H. V. Vu, P. L. Nguyen, Real-time hallucination correction in vision-language models using dynamic knowledge graph verification, Discover Artificial Intelligence (2026). doi:10.1007/s44163-026-01778-z.
URL https://doi.org/10.1007/s44163-026-01778-z

[11] V. Kotte, When can conformal risk control certify LLM outputs? bounds, impossibility, and adaptation for structured generation, arXiv preprint arXiv:2606.29054Version 2, 7 September 2026 (2026). doi:10.48550/arXiv.2606.29054.
URL https://arxiv.org/abs/2606.29054

[12] J. Wang, Y. Wang, G. Xu, J. Zhang, Y. Gu, H. Jia, J. Wang, H. Xu, M. Yan, J. Zhang, J. Sang, AMBER: An LLM-free multi-dimensional benchmark for MLLMs hallucination evaluation, arXiv preprint arXiv:2311.07397 (2023).
URL https://arxiv.org/abs/2311.07397

[13] Z. Bai, P. Wang, T. Xiao, T. He, Z. Han, Z. Zhang, M. Z. Shou, Hallucination of multimodal large language models: A survey, arXiv preprint arXiv:2404.18930 (2024).

[14] H. Liu, W. Xue, Y. Chen, D. Chen, X. Zhao, K. Wang, L. Hou, R. Li, W. Peng, A survey on hallucination in large vision–language models, arXiv preprint arXiv:2402.00253 (2024).

[15] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, ACM Computing Surveys 55 (12) (2023) 248:1–248:38.

[16] F. Liu, K. Lin, L. Li, J. Wang, Y. Yacoob, L. Wang, Mitigating hallucination in large multi-modal models via robust instruction tuning, in: International Conference on Learning Representations (ICLR), 2024.

[17] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: International Conference on Machine Learning (ICML), 2017.

[18] M. Minderer, A. A. Gritsenko, N. Houlsby, Scaling open-vocabulary object detection, in: Advances in Neural Information Processing Systems (NeurIPS), 2023.

[19] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, J. Zhu, L. Zhang, Grounding dino: Marrying dino with grounded pre-training for open-set object detection, in: European Conference on Computer Vision (ECCV), 2024.

[20] Y. Li, Y. Du, K. Zhou, J. Wang, W. X. Zhao, J.-R. Wen, Evaluating object hallucination in large vision-language models, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Singapore, 2023, pp. 292–305.

[21] C. Fu, P. Chen, Y. Shen, Y. Qin, M. Zhang, X. Lin, J. Yang, X. Zheng, K. Li, X. Sun, Y. Wu, R. Ji, C. Shan, R. He, MME: A comprehensive evaluation benchmark for multimodal large language models, in: Advances in Neural Information Processing Systems 38 (NeurIPS 2025),

Datasets and Benchmarks Track, Vol. 38, Curran Associates, Inc., 2025. doi:10.52202/085713-4899.
URL https://proceedings.neurips.cc/paper_files/paper/2025/file/d79a27cf2772fe00be7f341efc0eb517-Paper-Datasets_and_Benchmarks_Track.pdf

[22] D. A. Hudson, C. D. Manning, GQA: A new dataset for real-world visual reasoning and compositional question answering, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019.

[23] T. Guan, F. Liu, X. Wu, R. Xian, Z. Li, X. Liu, X. Wang, L. Chen, F. Huang, Y. Yacoob, D. Manocha, T. Zhou, HallusionBench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[24] Q. Huang, X. Dong, P. Zhang, B. Wang, C. He, J. Wang, D. Lin, W. Zhang, N. Yu, OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024.

[25] S. Leng, H. Zhang, G. Chen, X. Li, S. Lu, C. Miao, L. Bing, Mitigating object hallucinations in large vision-language models through visual contrastive decoding, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024.

[26] Z. Chen, Z. Zhao, H. Luo, H. Yao, B. Li, J. Zhou, Halc: Object hallucination reduction via adaptive focal-contrast decoding, in: International Conference on Machine Learning (ICML), 2024.

[27] A. Favero, L. Zancato, M. Trager, S. Choudhary, P. Perera, A. Achille, A. Swaminathan, S. Soatto, Multi-modal hallucination control by visual information grounding, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[28] C. Zhang, Z. Wan, Z. Kan, M. Q. Ma, S. Stepputtis, D. Ramanan, R. Salakhutdinov, L.-P. Morency, K. Sycara, Y. Xie, Self-correcting decoding with generative feedback for mitigating hallucinations in large vision-language models, in: International Conference on Learning Representations (ICLR), 2025.
URL https://arxiv.org/abs/2502.06130

[29] L. Zhao, Y. Deng, W. Zhang, Q. Gu, Mitigating object hallucination in large vision-language models via image-grounded guidance, in: Proceedings of the 42nd International Conference on Machine Learning (ICML), Vol. 267 of Proceedings of Machine Learning Research, PMLR, 2025, pp. 77461–77486, spotlight.
URL https://proceedings.mlr.press/v267/zhao25j.html

[30] Z. Jiang, J. Chen, B. Zhu, T. Luo, Y. Shen, X. Yang, Devils in middle layers of large vision-language models: Interpreting, detecting and mitigating object hallucinations via attention lens, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
URL https://arxiv.org/abs/2411.16724

[31] Z. Wan, C. Zhang, S. Yong, M. Q. Ma, S. Stepputtis, L.-P. Morency, D. Ramanan, K. Sycara, Y. Xie, ONLY: One-layer intervention sufficiently mitigates hallucinations in large vision-language models, in: IEEE/CVF International Conference on Computer Vision (ICCV), 2025, pp. 3225–3234.
URL https://arxiv.org/abs/2507.00898

[32] J. Deng, Y. Yang, MaskCD: Mitigating LVLM hallucinations by image head masked contrastive decoding, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Association for Computational Linguistics, 2025, pp. 18854–18866.
URL https://arxiv.org/abs/2510.02790

[33] Z. Cheng, W. Chen, F. Zhang, D. Shen, Mitigating hallucinations in large vision-language models via causal route gating, in: International Confer-

ence on Machine Learning (ICML), 2026, spotlight.
URL https://arxiv.org/abs/2605.24024

[34] Q. Zhao, X. Zhang, Y. Li, Y. Xing, X. Yuan, F. Tang, S. Fan, X. Chen, X. Zhang, D. Wang, MCA-LLaVA: Manhattan causal attention for reducing hallucination in large vision-language models, in: Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM), Association for Computing Machinery, 2025.
URL https://arxiv.org/abs/2507.09184

[35] Z. Sun, S. Shen, S. Cao, H. Liu, C. Li, Y. Shen, C. Gan, L. Gui, Y.-X. Wang, Y. Yang, K. Keutzer, T. Darrell, Aligning large multimodal models with factually augmented RLHF, in: Findings of the Association for Computational Linguistics: ACL 2024, Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 13088–13110.
URL https://aclanthology.org/2024.findings-acl.775/

[36] T. Yu, Y. Yao, H. Zhang, T. He, Y. Han, G. Cui, J. Hu, Z. Liu, H.-T. Zheng, M. Sun, T.-S. Chua, Rlhf-v: Towards trustworthy mllms via behavior alignment from fine-grained correctional human feedback, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[37] J. Lee, Y. Wang, J. Li, M. Zhang, Multimodal reasoning with multimodal knowledge graph, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), Bangkok, Thailand, 2024, pp. 10767–10782.
URL https://aclanthology.org/2024.acl-long.579/

[38] P. Xia, K. Zhu, H. Li, H. Zhu, Y. Li, G. Li, L. Zhang, H. Yao, RULE: Reliable multimodal RAG for factuality in medical vision language models, in: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), Miami, Florida, USA, 2024, pp. 1081–1093.

[39] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, New York, 2005.

[40] H. Papadopoulos, K. Proedrou, V. Vovk, A. Gammerman, Inductive confidence machines for regression, in: European Conference on Machine Learning (ECML), 2002, pp. 345–356.

[41] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, Journal of the American Statistical Association 113 (523) (2018) 1094–1111.

[42] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, Foundations and Trends in Machine Learning 16 (4) (2023) 494–591.

[43] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, Biometrika 26 (4) (1934) 404–413.

[44] I. Waudby-Smith, A. Ramdas, Estimating means of bounded random variables by betting, Journal of the Royal Statistical Society Series B 86 (1) (2024) 1–27.

[45] C. K. Chow, On optimum recognition error and reject tradeoff, IEEE Transactions on Information Theory 16 (1) (1970) 41–46.

[46] R. El-Yaniv, Y. Wiener, On the foundations of noise-free selective classification, Journal of Machine Learning Research 11 (2010) 1605–1641.

[47] P. L. Bartlett, M. H. Wegkamp, Classification with a reject option using a hinge loss, Journal of Machine Learning Research 9 (2008) 1823–1840.

[48] C. Cortes, G. DeSalvo, M. Mohri, Learning with rejection, in: International Conference on Algorithmic Learning Theory (ALT), 2016, pp. 67–82.

[49] Y. Geifman, R. El-Yaniv, Selectivenet: A deep neural network with an integrated reject option, in: International Conference on Machine Learning (ICML), 2019.

[50] V. Quach, A. Fisch, T. Schuster, A. Yala, J. H. Sohn, T. S. Jaakkola, R. Barzilay, Conformal language modeling, in: International Conference on Learning Representations (ICLR), 2024.
URL https://arxiv.org/abs/2306.10193

[51] C. Mohri, T. Hashimoto, Language models with conformal factuality guarantees, in: International Conference on Machine Learning (ICML), 2024, arXiv:2402.10978.

[52] M. Rubin-Toles, M. Gambhir, K. Ramji, A. Roth, S. Goel, Conformal language model reasoning with coherent factuality, in: International Conference on Learning Representations (ICLR), 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/679fcceef65c3d855aa885bd024542c1-Paper-Conference.pdf

[53] A. Rubashevskii, D. Piatrashyn, P. Nakov, M. Panov, Adaptive conformal prediction for improving factuality of generations by large language models, arXiv preprint arXiv:2604.13991 (2026).
URL https://arxiv.org/abs/2604.13991

[54] Y. Chen, D. Chen, S. M. Chikodikar, C. H. Yin, R. K. Vinayak, Is conformal factuality for RAG-based LLMs robust? novel metrics and systematic insights, arXiv preprint arXiv:2603.16817 (2026).
URL https://arxiv.org/abs/2603.16817

[55] Y. Gui, Y. Jin, Z. Ren, Conformal alignment: Knowing when to trust foundation models with guarantees, in: Advances in Neural Information Processing Systems (NeurIPS), 2024.

[56] B. Kumar, C. Lu, G. Gupta, A. Palepu, D. Bellamy, R. Raskar, A. Beam, Conformal prediction with large language models for multi-choice question answering, in: ICML 2023 Workshop on Neural Conversational AI: What's Left to TEACH (Trustworthy, Enhanced, Adaptable, Capable and Human-centric) Chatbots?, 2023.
URL https://arxiv.org/abs/2305.18404

[57] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh, S. Tu, N. Brown, P. Xu, L. Takayama, F. Xia, J. Varley, Z. Xu, D. Sadigh, A. Zeng, A. Majumdar, Robots that ask for help: Uncertainty alignment for large language model planners, in: Conference on Robot Learning (CoRL), 2023.

[58] B. J. Wagner, Two axes of LLM abstention: Answer correctness and question answerability, arXiv preprint arXiv:2607.08456 (2026).

[59] T. Wang, Y. Shi, Y. Yan, H. Zhang, Inference-time conformal reasoning with valid factuality control for large language models, arXiv preprint arXiv:2606.08831ArXiv comment: Accepted at ICML 2026 (proceedings entry not checked) (2026).
URL https://arxiv.org/abs/2606.08831

[60] Z. Jiang, A. Liu, B. Van Durme, Conformal linguistic calibration: Trading-off between factuality and specificity, in: Advances in Neural Information Processing Systems (NeurIPS), main conference, 2025, arXiv:2502.19110. Proceedings volume and pages not checked.
URL https://arxiv.org/abs/2502.19110

[61] D. Madras, T. Pitassi, R. Zemel, Predict responsibly: Improving fairness and accuracy by learning to defer, in: Advances in Neural Information Processing Systems, Vol. 31, 2018.
URL https://proceedings.neurips.cc/paper_files/paper/2018/file/09d37c08f7b129e96277388757530c72-Paper.pdf

[62] H. Mozannar, D. Sontag, Consistent estimators for learning to defer to an expert, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 7076–7087.
URL https://proceedings.mlr.press/v119/mozannar20b.html

[63] L. Chen, M. Zaharia, J. Zou, FrugalGPT: How to use large language models while reducing cost and improving performance, Transactions on Machine

Learning Research (2024). arXiv:2305.05176.
URL https://openreview.net/forum?id=cSimKw5p6R

[64] J. Jung, F. Brahman, Y. Choi, Trust or escalate: LLM judges with provable guarantees for human agreement, in: International Conference on Learning Representations (ICLR), 2025. arXiv:2407.18370.
URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/08dabd5345b37fffcbe335bd578b15a0-Abstract-Conference.html

[65] Y. Dou, S. Lian, S. Li, Conformal cascade: Distribution-free accuracy guarantees for multi-tier LLM inference, arXiv preprint (2026). arXiv:2607.25018.

[66] J. Huang, S. Park, N. Paoletti, O. Simeone, Reliable inference in edge-cloud model cascades via conformal alignment, arXiv preprint (2025). arXiv:2510.17543.

[67] W. Overman, M. Bayati, Conformal arbitrage: Risk-controlled balancing of competing objectives in language models, arXiv preprint (2025). arXiv:2506.00911.

[68] T. Bary, B. Macq, L. Petit, No need for learning to defer? a training free deferral framework to multiple experts through conformal prediction, arXiv preprint (2025). arXiv:2509.12573.

[69] S. Kadavath, T. Conerly, A. Askell, T. Henighan, et al., Language models (mostly) know what they know, arXiv preprint arXiv:2207.05221 (2022).

[70] L. Kuhn, Y. Gal, S. Farquhar, Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, in: International Conference on Learning Representations (ICLR), 2023.

[71] S. Farquhar, J. Kossen, L. Kuhn, Y. Gal, Detecting hallucinations in large language models using semantic entropy, Nature 630 (2024) 625–630.

[72] S. Lin, J. Hilton, O. Evans, Teaching models to express their uncertainty in words, Transactions on Machine Learning Research (2022).

[73] P. Manakul, A. Liusie, M. J. F. Gales, Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models, in: Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

[74] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations (ICLR), 2023.

[75] B. Xu, Y. Lu, TECP: Token-entropy conformal prediction for LLMs, Mathematics 13 (20) (2025) 3351.
URL https://doi.org/10.3390/math13203351

[76] Y. Chen, Y. Wang, S. Liu, Y. Jing, D. Tao, CoVeR: Conformal calibration for versatile and reliable autoregressive next-token prediction, arXiv preprint arXiv:2509.04733 (2025).

[77] A. Mishra, T. Kumar, A. Shah, S. Bhattacharya, M. Foltin, CEBC: Conformal evidence-bounded control for low-hallucination vision–language generation, in: M. Liakata, V. P. Moreira, J. Zhang, D. Jurgens (Eds.), Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, San Diego, California, United States, 2026, pp. 46193–46206. doi:10.18653/v1/2026.acl-long.2142.
URL https://aclanthology.org/2026.acl-long.2142/

[78] C. Fannjiang, S. Bates, A. N. Angelopoulos, J. Listgarten, M. I. Jordan, Conformal prediction under feedback covariate shift for biomolecular design, Proceedings of the National Academy of Sciences 119 (43) (2022).

[79] R. J. Tibshirani, R. F. Barber, E. J. Candès, A. Ramdas, Conformal prediction under covariate shift, in: Advances in Neural Information Processing Systems (NeurIPS), 2019, pp. 2526–2536.

[80] I. Gibbs, E. J. Candès, Adaptive conformal inference under distribution shift, in: Advances in Neural Information Processing Systems (NeurIPS), 2021.

[81] M. Lee, Y. Jung, S. Park, Online conformal abstention for factuality control under adversarial bandit feedback, arXiv preprint arXiv:2506.14067 (2025).

[82] D. Prinster, S. Stanton, A. Liu, S. Saria, Conformal validity guarantees exist for any data distribution (and how to find them), in: Proceedings of the 41st International Conference on Machine Learning (ICML), Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 41086–41118.
URL https://arxiv.org/abs/2405.06627

[83] H. Liu, C. Li, Y. Li, Y. J. Lee, Improved baselines with visual instruction tuning, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024.

[84] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, J. Lin, Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution, arXiv preprint arXiv:2409.12191 (2024).

[85] A. Ramdas, P. Grünwald, V. Vovk, G. Shafer, Game-theoretic statistics and safe anytime-valid inference, Statistical Science 38 (4) (2023) 576–601.

Table A.16: Certified coverage (%) with abort rate (ab) and validity statistics for every arm, all items. exc is the fraction of non-aborted splits whose test-fold risk exceeds α; cv is the fraction whose one-sided 90% Clopper–Pearson lower bound on test risk exceeds α. The fixed q=0.10 gate was chosen with knowledge of these benchmarks and is not certified.

		Filter		External, size-matched		External, full-size		Fit-fold gate		Union ($\delta/3$)		Fixed q=0.10	
Setting	α	cov. (ab)	exc/cv	cov. (ab)	exc/cv	cov. (ab)	exc/cv	cov. (ab)	exc/cv	cov. (ab)	exc/cv	cov. (ab)	exc/cv
POPE-1500, LLaVA	0.10	68.2 (0)	0.11 / 0.02	69.6 (0)	0.11 / 0.01	77.6 (0)	0.09 / 0.01	77.0 (1)	0.10 / 0.01	72.6 (0)	0.04 / 0.01	71.1 (0)	0.11 / 0.01
	0.15	85.5 (0)	0.09 / 0.01	92.8 (0)	0.09 / 0.01	92.8 (0)	0.09 / 0.01	92.6 (0)	0.09 / 0.01	89.1 (0)	0.04 / 0.00	87.6 (0)	0.09 / 0.01
POPE-3000, LLaVA	0.10	76.6 (0)	0.06 / 0.01	83.2 (0)	0.09 / 0.01	83.2 (0)	0.09 / 0.01	82.8 (0)	0.09 / 0.01	79.3 (0)	0.03 / 0.01	78.6 (0)	0.07 / 0.01
	0.15	90.8 (0)	0.11 / 0.01	95.8 (0)	0.10 / 0.01	95.8 (0)	0.10 / 0.01	95.8 (0)	0.10 / 0.01	94.0 (0)	0.04 / 0.00	92.2 (0)	0.10 / 0.01
POPE-adv, LLaVA	0.10	44.5 (25)	0.14 / 0.01	46.2 (24)	0.14 / 0.01	37.3 (48)	0.13 / 0.01	41.4 (34)	0.14 / 0.00	28.0 (57)	0.12 / 0.01	48.2 (22)	0.14 / 0.02
	0.15	72.5 (2)	0.07 / 0.00	74.4 (1)	0.08 / 0.00	82.6 (2)	0.07 / 0.00	82.0 (2)	0.08 / 0.00	72.7 (4)	0.04 / 0.00	76.4 (1)	0.09 / 0.00
POPE-adv, Qwen2-VL	0.10	76.9 (1)	0.09 / 0.01	78.0 (0)	0.10 / 0.01	69.1 (14)	0.11 / 0.01	71.2 (10)	0.11 / 0.01	68.9 (11)	0.05 / 0.00	79.4 (0)	0.10 / 0.01
	0.15	94.6 (0)	0.06 / 0.00	95.7 (0)	0.06 / 0.00	95.7 (0)	0.06 / 0.00	95.0 (0)	0.06 / 0.00	92.9 (0)	0.04 / 0.00	95.2 (0)	0.06 / 0.00
POPE-adv, LLaVA+VCD	0.10	45.8 (14)	0.14 / 0.01	46.6 (12)	0.13 / 0.01	33.5 (46)	0.23 / 0.04	39.6 (30)	0.21 / 0.03	34.2 (37)	0.10 / 0.01	47.8 (11)	0.14 / 0.01
	0.15	74.4 (0)	0.11 / 0.02	76.8 (0)	0.11 / 0.02	76.8 (0)	0.11 / 0.02	76.1 (0)	0.11 / 0.02	69.4 (0)	0.04 / 0.01	75.7 (0)	0.11 / 0.02
AMBER(d), LLaVA	0.10	64.4 (0)	0.10 / 0.01	67.5 (0)	0.09 / 0.01	58.9 (13)	0.10 / 0.02	66.8 (1)	0.10 / 0.01	66.4 (0)	0.06 / 0.01	64.4 (5)	0.09 / 0.02
	0.15	75.7 (0)	0.12 / 0.02	79.6 (0)	0.13 / 0.01	79.6 (0)	0.13 / 0.01	79.5 (0)	0.13 / 0.02	77.7 (0)	0.06 / 0.01	78.8 (0)	0.13 / 0.02

Table A.17: As Table 4 at $\alpha = 0.15$.

	POPE-1500, LLaVA	POPE-3000, LLaVA	POPE-adv, LLaVA	POPE-adv, Qwen2-VL	POPE-adv, LLaVA+VCD	AMBER(d), LLaVA
conventional protocol (item folds, calibration quantiles, e_0=0, fixed q=0.10)	86.4/88.4	91.0/92.4	69.3/76.7	94.8/95.2	75.8/77.2	96.2/59.9
+ image-grouped folds	85.8/87.8	91.0/92.3	70.4/77.7	94.5/94.9	75.6/76.8	95.9/64.7
+ fixed value grid and gate (fit fold)	85.5/87.5	90.8/92.2	61.8/72.9	93.7/94.7	63.1/72.4	85.1/85.0
+ analytic start e_0=1	85.5/87.6	90.8/92.2	70.9/75.1	94.1/94.9	73.5/75.1	94.2/94.1
+ gate chosen on the fit fold	85.5/92.6	90.8/95.8	70.9/81.1	94.1/94.7	73.5/75.5	94.2/96.4
+ all-item cohort	85.5/92.6	90.8/95.8	72.5/82.0	94.6/95.0	74.4/76.1	75.7/79.5

Table A.18: As Table 7 at $\alpha = 0.05$.

			POPE-1500, LLaVA		POPE-3000, LLaVA		POPE-adv, LLaVA		POPE-adv, Qwen2-VL		POPE-adv, LLaVA+VCD		AMBER(d), LLaVA	
c	e_0	k_{start}	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC
1.0	0	45	19.6 (35)	29.2 (18)	47.0 (2)	46.9 (9)	8.6 (80)	7.9 (82)	35.4 (17)	36.7 (19)	9.8 (74)	9.6 (75)	48.0 (0)	47.2 (5)
1.0	1	77	27.5 (14)	31.4 (16)	47.2 (1)	46.9 (10)	3.2 (94)	3.2 (94)	31.6 (46)	29.6 (50)	8.7 (82)	8.7 (82)	48.2 (0)	47.8 (6)
1.0	2	105	24.2 (32)	27.2 (30)	47.8 (0)	46.4 (10)	0.5 (99)	0.5 (99)	37.1 (40)	34.3 (46)	2.2 (96)	2.2 (96)	48.3 (0)	47.7 (6)
1.0	3	132	21.2 (50)	22.2 (51)	47.8 (0)	45.5 (12)	0.0 (100)	0.0 (100)	7.4 (90)	7.3 (90)	0.3 (100)	0.3 (100)	48.2 (0)	46.3 (9)
1.5	0	45	26.3 (16)	31.6 (15)	47.3 (1)	46.5 (9)	4.7 (91)	4.6 (91)	30.4 (44)	26.6 (52)	8.2 (82)	8.2 (82)	48.1 (0)	47.4 (6)
1.5	1	77	21.8 (44)	24.3 (41)	47.9 (0)	46.5 (10)	0.0 (100)	0.0 (100)	24.1 (62)	23.4 (64)	1.2 (98)	1.2 (98)	48.2 (0)	47.8 (6)
1.5	2	105	23.1 (50)	24.6 (48)	47.9 (0)	45.2 (12)	0.0 (100)	0.0 (100)	1.6 (98)	1.6 (98)	0.0 (100)	0.0 (100)	48.3 (0)	45.2 (11)
1.5	3	132	25.0 (49)	25.8 (48)	47.7 (0)	45.5 (12)	0.0 (100)	0.0 (100)	0.0 (100)	0.0 (100)	0.0 (100)	0.0 (100)	48.4 (0)	46.2 (9)

Table A.19: As Table 7 at $\alpha = 0.15$.

			POPE-1500, LLaVA		POPE-3000, LLaVA		POPE-adv, LLaVA		POPE-adv, Qwen2-VL		POPE-adv, LLaVA+VCD		AMBER(d), LLaVA	
c	e_0	k_{start}	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC	filter	CCRC
1.0	0	15	85.5 (0)	92.6 (0)	90.8 (0)	95.8 (0)	46.1 (36)	81.9 (1)	91.9 (3)	92.6 (2)	38.9 (47)	73.4 (4)	75.7 (0)	79.5 (0)
1.0	1	25	85.5 (0)	92.6 (0)	90.8 (0)	95.8 (0)	67.6 (8)	82.9 (0)	94.3 (0)	94.9 (0)	66.2 (11)	74.7 (2)	75.7 (0)	79.5 (0)
1.0	2	34	85.6 (0)	92.6 (0)	90.8 (0)	95.8 (0)	71.4 (3)	82.4 (1)	94.3 (0)	94.8 (0)	73.2 (2)	75.2 (1)	75.7 (0)	79.5 (0)
1.0	3	43	85.6 (0)	92.6 (0)	90.8 (0)	95.8 (0)	73.2 (1)	81.3 (2)	94.5 (0)	94.9 (0)	74.4 (0)	75.6 (0)	75.7 (0)	79.5 (0)
1.5	0	15	85.5 (0)	92.6 (0)	90.8 (0)	95.8 (0)	64.0 (12)	82.4 (0)	94.3 (0)	94.8 (0)	61.7 (18)	74.0 (3)	75.7 (0)	79.5 (0)
1.5	1	25	85.5 (0)	92.6 (0)	90.8 (0)	95.8 (0)	72.5 (2)	82.0 (2)	94.6 (0)	95.0 (0)	74.4 (0)	76.1 (0)	75.7 (0)	79.5 (0)
1.5	2	34	85.5 (0)	92.6 (0)	90.8 (0)	95.8 (0)	72.6 (3)	81.3 (3)	94.6 (0)	95.2 (0)	74.5 (0)	76.1 (0)	75.7 (0)	79.5 (0)
1.5	3	43	85.6 (0)	92.6 (0)	90.8 (0)	95.8 (0)	72.8 (4)	79.9 (5)	94.8 (0)	95.1 (0)	74.6 (1)	74.6 (3)	75.6 (0)	79.4 (0)

Appendix A. Additional results

Appendix A.1. Audit of all arms

Appendix A.2. Protocol ladder and start of the sequence at other targets

Appendix A.3. Gain decomposition

Appendix A.4. Multi-start search

Appendix A.5. Benchmarks where the detector does not apply

Appendix A.6. Grounded-only cohort 54

Appendix A.7. Reproducibility

All numbers in the paper are generated from per-item signals by the scripts in the released repository (`conformal_followup/revision`): `attach_image_ids.py`

Table A.20: Gain decomposition (points of test coverage), $q = 0.10$, all items, conditional on both arms certifying.

Setting	α	threshold moves (%)	gain	(i) repairs at filter's threshold	(ii) dilution [95% CI over splits]
POPE-1500, LLaVA	0.10	31	2.79	1.89	0.89 [0.74, 1.05]
	0.15	25	2.06	1.43	0.63 [0.51, 0.75]
POPE-3000, LLaVA	0.10	24	2.02	1.46	0.56 [0.45, 0.67]
	0.15	13	1.42	1.12	0.30 [0.22, 0.38]
POPE-adv, LLaVA	0.10	24	2.75	1.94	0.81 [0.52, 1.10]
	0.15	26	3.04	1.59	1.46 [0.99, 1.92]
POPE-adv, Qwen2-VL	0.10	18	2.17	0.93	1.24 [0.64, 1.84]
	0.15	3	0.37	0.31	0.06 [0.01, 0.10]
POPE-adv, LLaVA+VCD	0.10	12	1.12	0.67	0.45 [0.25, 0.66]
	0.15	11	1.14	0.49	0.65 [0.28, 1.02]
AMBER(d), LLaVA	0.10	24	3.39	3.24	0.15 [0.03, 0.27]
	0.15	34	3.33	2.58	0.75 [0.64, 0.86]

Table A.21: Certified coverage (%, abort rate in parentheses) for the single starts $e_0 = 0$ and $e_0 = 1$ and for the union over $e_0 \in \{0, 1, 2\}$ at $\delta/3$ each.

		filtering			CCRC, external gate		
Setting	α	e_0=0	e_0=1	multi-start ($\delta/3$)	e_0=0	e_0=1	multi-start ($\delta/3$)
POPE-1500, LLaVA	0.10	67.5 (0)	68.2 (0)	64.2 (0)	69.5 (0)	69.6 (0)	66.3 (0)
	0.15	85.5 (0)	85.5 (0)	83.2 (0)	92.8 (0)	92.8 (0)	91.2 (0)
POPE-3000, LLaVA	0.10	76.5 (0)	76.6 (0)	74.4 (0)	83.2 (0)	83.2 (0)	81.2 (0)
	0.15	90.8 (0)	90.8 (0)	89.2 (0)	95.8 (0)	95.8 (0)	95.1 (0)
POPE-adv, LLaVA	0.10	36.9 (30)	44.5 (25)	31.9 (41)	39.3 (27)	46.2 (24)	33.2 (40)
	0.15	64.0 (12)	72.5 (2)	65.1 (3)	68.2 (8)	74.4 (1)	66.5 (2)
POPE-adv, Qwen2-VL	0.10	76.5 (0)	76.9 (1)	72.4 (1)	78.0 (0)	78.0 (0)	73.4 (1)
	0.15	94.3 (0)	94.6 (0)	91.8 (0)	95.5 (0)	95.7 (0)	92.8 (0)
POPE-adv, LLaVA+VCD	0.10	39.5 (22)	45.8 (14)	35.4 (28)	40.8 (20)	46.6 (12)	36.2 (26)
	0.15	61.7 (18)	74.4 (0)	64.8 (2)	76.5 (1)	76.8 (0)	66.9 (5)
AMBER(d), LLaVA	0.10	64.4 (0)	64.4 (0)	63.4 (0)	67.5 (0)	67.5 (0)	65.4 (2)
	0.15	75.7 (0)	75.7 (0)	74.2 (0)	79.6 (0)	79.6 (0)	78.1 (0)

recovers and verifies image identifiers; `ccrc_rev.py` implements the protocol; `regress_legacy.py` checks that the conventional-protocol mode reproduces the numbers of the baseline implementation; `exp_*.py` produce the results files and `make_tables.py`, `make_figures.py` the tables and figures. Splits are deterministic in the seed.

Table A.22: Filtering (confidence score, image-grouped folds, 400 splits) where the detector does not apply. Certified coverage (%) with abort rate in parentheses.

Dataset	images	μ (%)	detector value	α=0.10	α=0.15	α=0.20
MME (700)	350	31.1	0	0.9 (97)	10.4 (56)	21.0 (24)
GQA testdev (700)	319	27.6	0	8.5 (72)	34.6 (18)	57.7 (4)
HallusionBench (447)	165	48.8	0	0.0 (100)	0.0 (100)	0.0 (100)

Table A.23: As Table 3 restricted to items with a detector value; no bootstrap.

		Filter	External gate		Fit-fold gate		Fixed q=0.10	
Setting	α	cov. (ab)	cov. (ab)	gain	cov. (ab)	gain	cov. (ab)	gain
POPE-1500, LLaVA	0.10	68.2 (0)	69.6 (0)	+1.4	77.0 (1)	+8.8	71.1 (0)	+2.9
	0.15	85.5 (0)	92.8 (0)	+7.3	92.6 (0)	+7.1	87.6 (0)	+2.1
POPE-3000, LLaVA	0.10	76.6 (0)	83.2 (0)	+6.6	82.8 (0)	+6.3	78.6 (0)	+2.0
	0.15	90.8 (0)	95.8 (0)	+5.1	95.8 (0)	+5.0	92.2 (0)	+1.4
POPE-adv, LLaVA	0.10	43.6 (26)	45.5 (24)	+2.0	39.4 (37)	−4.2	47.4 (22)	+3.8
	0.15	70.9 (2)	72.6 (2)	+1.6	81.1 (1)	+10.2	75.1 (1)	+4.2
POPE-adv, Qwen2-VL	0.10	75.6 (1)	76.9 (1)	+1.3	68.3 (13)	−7.3	78.0 (1)	+2.5
	0.15	94.1 (0)	95.3 (0)	+1.2	94.7 (0)	+0.6	94.9 (0)	+0.8
POPE-adv, LLaVA+VCD	0.10	45.5 (15)	46.1 (14)	+0.6	39.7 (30)	−5.8	47.6 (12)	+2.1
	0.15	73.5 (2)	77.2 (1)	+3.7	75.5 (2)	+2.0	75.1 (1)	+1.6
AMBER(d), LLaVA	0.10	90.9 (0)	93.6 (0)	+2.7	93.3 (0)	+2.4	90.1 (2)	−0.8
	0.15	94.2 (1)	96.5 (0)	+2.3	96.4 (0)	+2.2	94.1 (2)	−0.1